Time for a Quick Start to Time Series Analysis

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#MWSUG2026

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Abstract

Never had the TIME to investigate time series analysis? This is the quick start talk for you! This presentation will allow participants to start with a software agnostic introduction to time series where we will answer "What is a Time Series?" and "What are components of a Time Series?". We will then move into a discussion of the three main classes of time series models (ESM, UCM, and ARIMAX). Remaining time will allow the participant to see modeling of time series structures in both SAS 9.4 and in SAS Viya Visual Forecasting. Within SAS 9.4, emphasis will be placed on utilizing the SAS Studio tasks that will prompt the user for choices in their analysis. Time will be taken to view the associated code that is generated in SAS 9.4. This presentation will also take time to show SAS Viya Model Studio in which users can perform time series analysis in another GUI aspect. Discussion will be had concerning differences and similarities between the SAS 9.4 options and the SAS Visual Forecasting options. Do you have the TIME for this Quick Start?

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Time Series Elements

Time Series Characteristics

Objectives

- Define a time series and the main ideas in time series data creation.
- Use the TIMESERIES procedure to accumulate transactional data into time series data.
- Define and explore the systematic components in a time series.
- Identify the accuracy statistics that are used to measure forecast accuracy.
- Describe three types of time series models that are discussed in this course.

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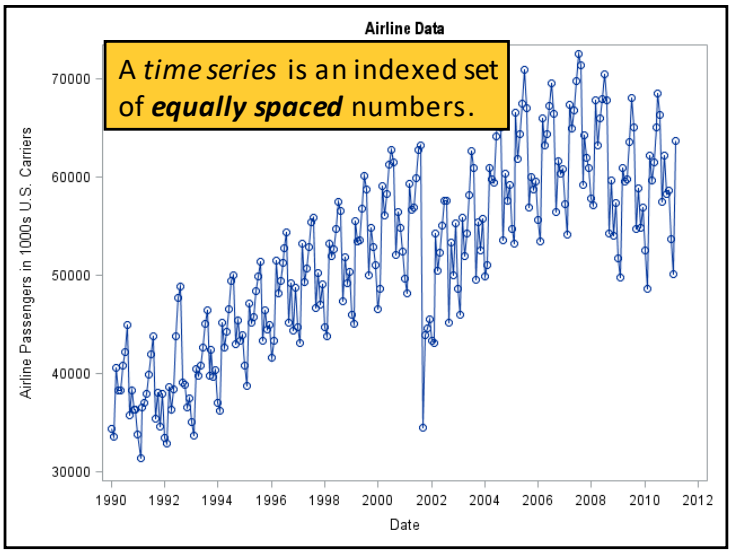
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In business applications, time series usually start as transactional or timestamped data. An example of transactional data is a record of customer visits to a website over a period of a year. Each visit is recorded with a customer identifier and a timestamp.

Transactional data are not organized with respect to a time interval. They must be made equally spaced, or indexed, before time series models can be used to quantify the systematic variation contained in it. *Accumulation* is the process of indexing or transforming transactional data into time series.

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A Time Series



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Time Series Creation

Data Accumulation:

Accumulates transactional data to a specified time interval of the data.

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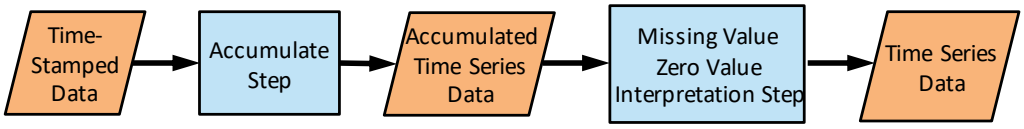
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Transactional Data Preparation Steps

Before you can perform time series analysis and forecasting, the observation (raw) series must be accumulated and interpreted to form a time series.

The following diagram summarizes the process of converting timestamped data to time series data:



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The selection of the interval and the accumulation method are critical considerations in applied time series modeling. When making these choices, the modeler ***creates*** the data for analysis. Different accumulation choices can emphasize different systematic characteristics of the data, and impact model usefulness and precision.

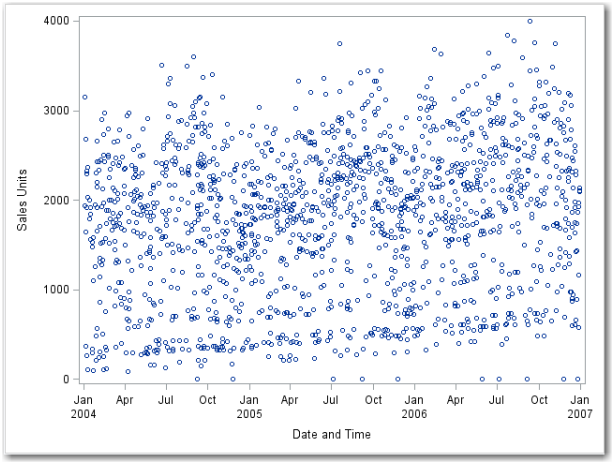
For example, consider timestamped calls into a call center. Accumulating the data to an hourly interval using a sum accumulation method gives analysts, assuming adequate volume, a good look at the hour-of-the-day cycle. It also provides direct information about what the peak and trough hours for calls are in a 24-hour cycle. This information is helpful for daily staffing decisions.

However, time series accumulated at an hourly interval might not be optimal for understanding and quantifying longer term trends and month-of-the-year cycles that might exist in the data. A better alternative for understanding and quantifying longer run patterns might be a monthly interval and an average accumulation method.

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Transactional Data: Example

Historical Timestamped Data



No obvious patterns exist in the transactional data.

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Transactional Analysis

- Given a timestamped data set, each observation of the data set can be assigned an observation (raw) index, a time index, and a season index.
- Count and frequency analysis can be applied to the timestamped data set based on these indices.
- Each of these indices does not depend on the data under analysis.
- The indices only structure the data for subsequent analysis.

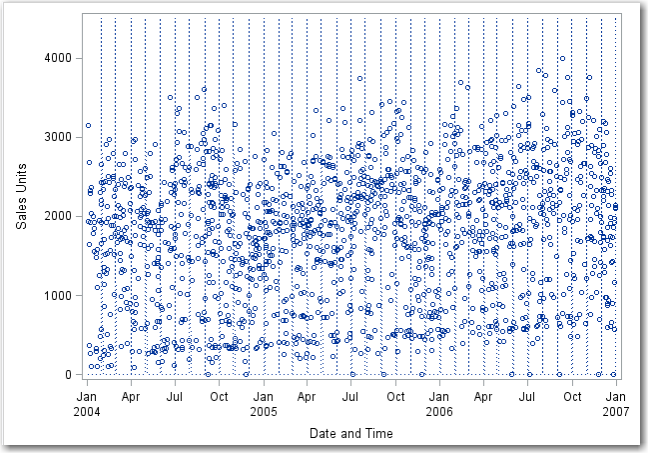
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Transactional Data Time Binning

Monthly Time Bins



A monthly interval time series has one observation per interval or bin.

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Transactional Data Time Binning

Perform time binning with the TIMESERIES procedure's INTERVAL= option.
Use accumulated totals.

```
proc timeseries data=transactions out=outsum;
   id date interval=month
           accumulate=total;
   var units;
run;
```

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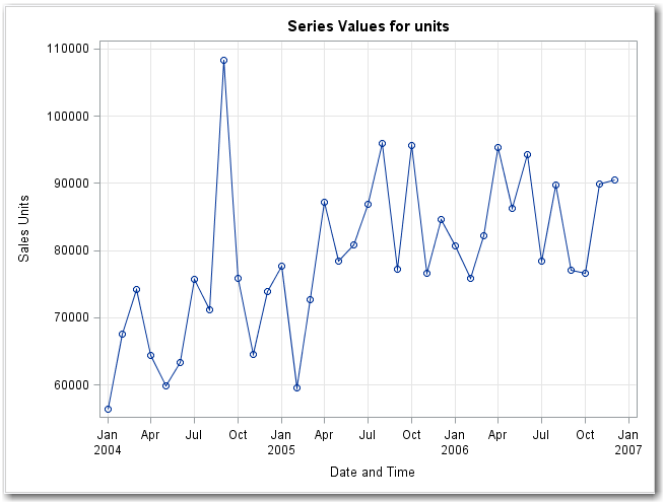
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Accumulation Options

- **NONE**: No accumulation occurs. The ID variable values must be equally spaced with respect to the frequency. This is the default option.
- **TOTAL**: Observations are accumulated based on the total sum of their values.
- **AVERAGE** | **AVG**: Observations are accumulated based on the average of their values.
- **MINIMUM** | **MIN**: Observations are accumulated based on the minimum of their values.
- **MEDIAN** | **MED**: Observations are accumulated based on the median of their values.
- **MAXIMUM** | **MAX**: Observations are accumulated based on the maximum of their values.
- **N**: Observations are accumulated based on the number of nonmissing observations.
- **NMISS**: Observations are accumulated based on the number of missing observations.
- **NOBS**: Observations are accumulated based on the number of observations.
- **FIRST**: Observations are accumulated based on the first of their values.
- **LAST**: Observations are accumulated based on the last of their values.
- **STDDEV** |**STD**: Observations are accumulated based on the standard deviation of their values.
- **CSS**: Observations are accumulated based on the corrected sum of squares of their values.
- **USS**: Observations are accumulated based on the uncorrected sum of squares of their values.

Transactional Data Accumulation

Accumulated on a Monthly ***Total*** Basis



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Transactional Data Time Binning

Perform time binning with the TIMESERIES procedure's INTERVAL= option. Use accumulated averages.

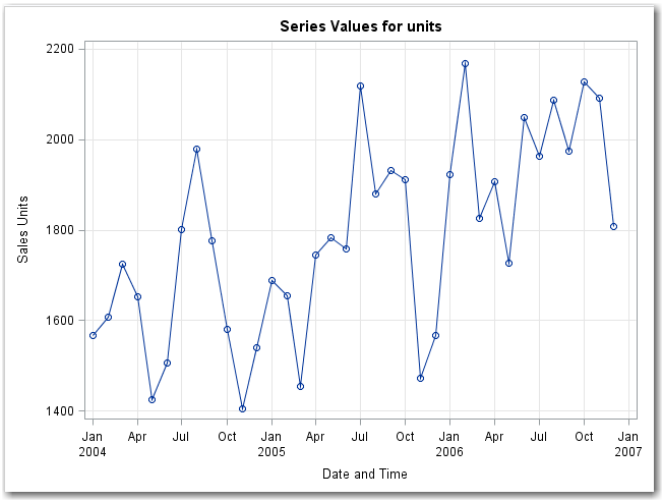
```
proc timeseries data=transactions out=outavg;
   id interval=month
      accumulate=average;
   var units;
run;
```

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Transactional Data Accumulation

Accumulated on a Monthly ***Average*** Basis



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Transactional Data Time Binning

Perform time binning with the TIMESERIES procedure's INTERVAL= option. Use the maximum value in each interval.

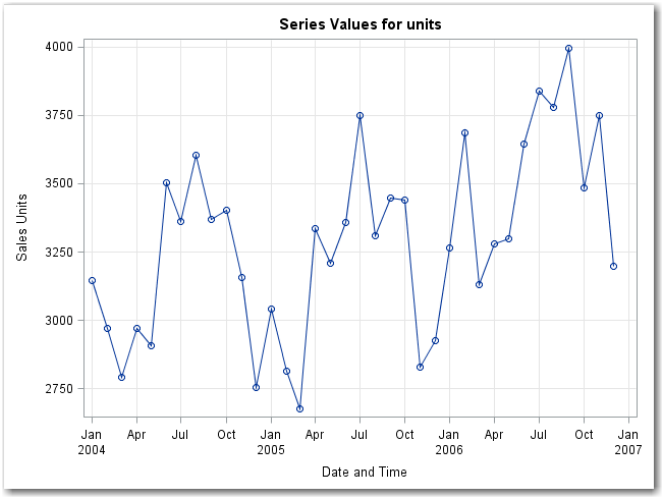
```
proc timeseries data=transactions out=outmax;
   id interval=month
      accumulate=maximum;
   var units;
run;
```

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Transactional Data Accumulation

Accumulated on a Monthly **Maximum** Basis



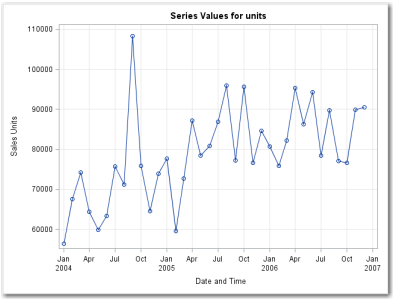
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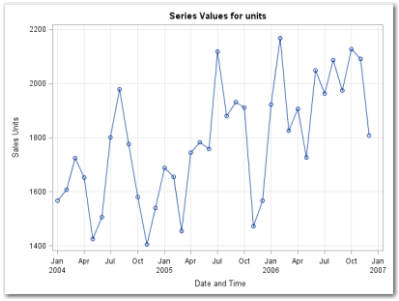
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Transactional Data Accumulation

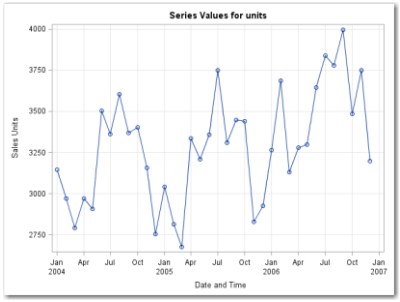
For this timestamped data, accumulating on a monthly ***average*** or ***maximum*** basis might be better than accumulating on a ***total*** basis.



ACCUMULATE=TOTAL



ACCUMULATE=AVERAGE



ACCUMULATE=MAXIMUM

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Time Series Components

Objectives

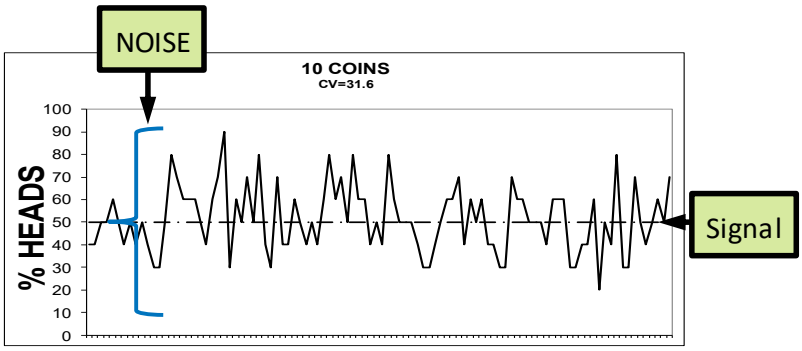
- Describe the decomposition of time series variation.
- List the main components of systematic variation or signal that exists in time series.
- Perform a preliminary identification of a time series.

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Variation in Time Series Data: Two Main Parts

- signal
- noise

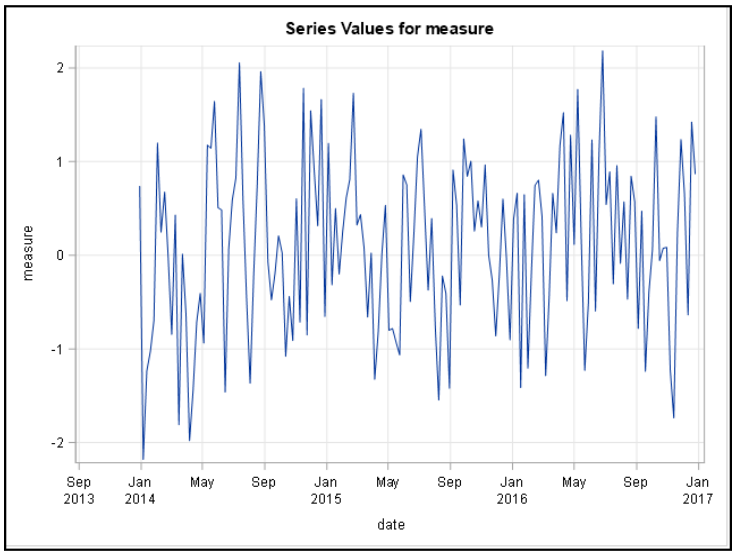


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White Noise



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A time series that consists of only noise is often referred to as a *white noise series*. A white noise series is a series of independently, identically distributed (i.i.d.) random variables. Each distribution has a mean of 0 and the same finite variance. A white noise series (from an underlying "white noise process") contains, by definition, no signal. The forecasted value for such a series, at any time point, does not change, and it is generally calculated to be zero. Where there is no signal, there are no patterns to help produce a better forecast than that.

Signal Components

- level
- trend
- seasonality
- cycle
- exogenous (also known as explanatory variable effects)
- irregular

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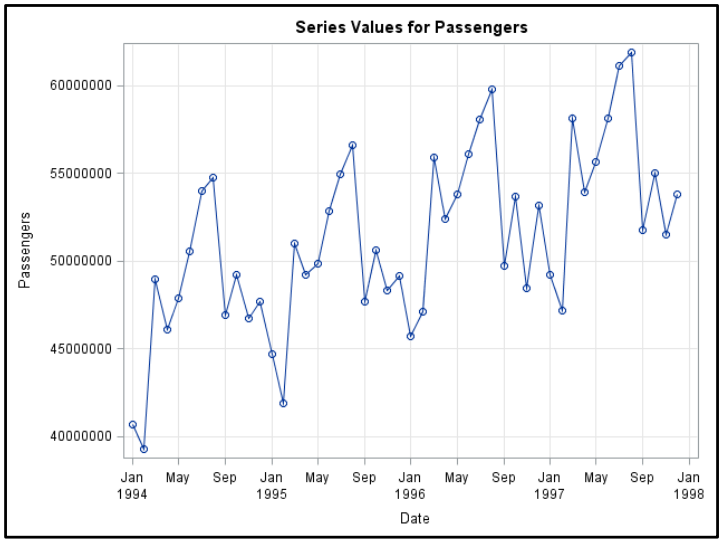
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Signal can be decomposed into several types.

- *Level* can be thought of as the baseline mean at any time point. It is not restricted to the value *zero*.
- *Trend* is the long-term pattern of the mean of a series. It can be linear, quadratic, or more complex. The concept of trend is typically involved with the level of a series. This is analogous to linear regression, where the straight-line model includes both the linear trend (the slope) and the base level (the intercept).
- *Seasonality* can be considered the predictable deviations of the series patterns from the trend component. The pattern of the deviations has a regular period (for example, 52 for weekly series, 12 for monthly series, and so on). The deviations from the series might or might not consist of a smooth (sinusoidal) cycle.
- The *cycle* component, in contrast with the seasonal component, is a regular, repetitive cycle that does not have a regular period. For example, one cycle might be a duration of 20 months and the next, 25 months.
- *Exogenous* effects are not really components of the series. However, they are sources of variation in the series that might be due to, or related with, the values of another series. They are sometimes known as *explanatory effects*. That relationship might be delayed or concurrent.
- Finally, the *irregular* component is what is remains after the patterns for all other effects and components are accounted for. It is the residual variation. The irregular component can be white noise, but that is not required. There can be local, forecastable patterns in the residuals, such as autocorrelation and moving average (discussed later).

The Airline Data



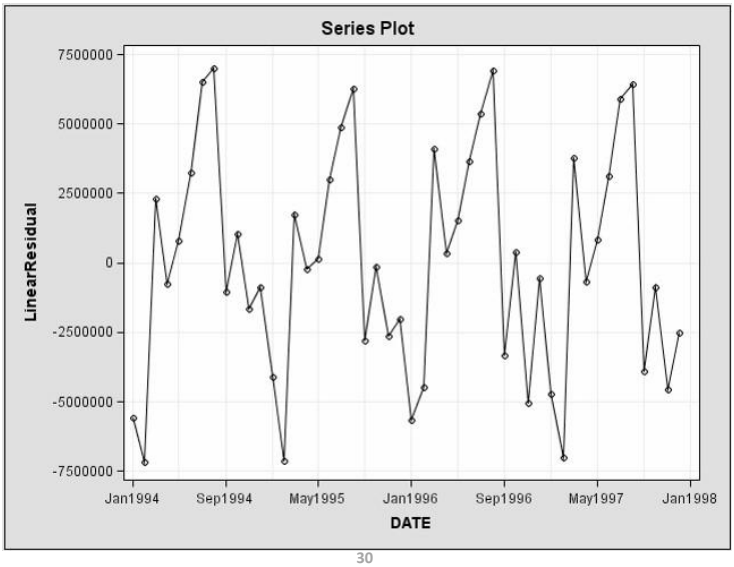
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The airline data, **DOTAIR9497**, contain many of the signal components listed above. The data set contains a monthly, intervaled time series of passengers who flew on commercial aircraft in the United States between January 1994 and December 1997. The slides below show a decomposition of the series into seasonal, trend, and irregular components.

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Signal Components: Seasonality

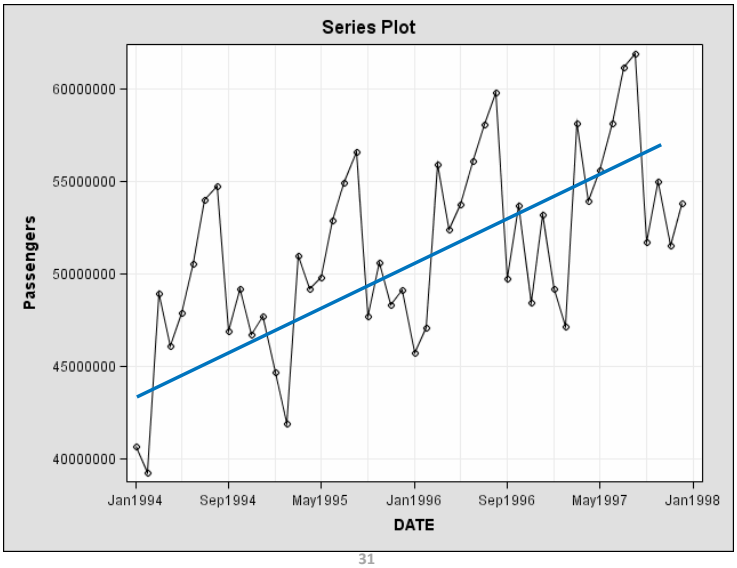


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Signal Components: Trend



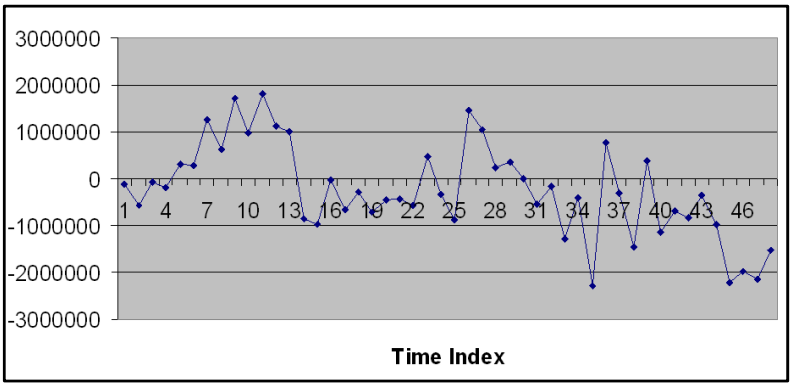
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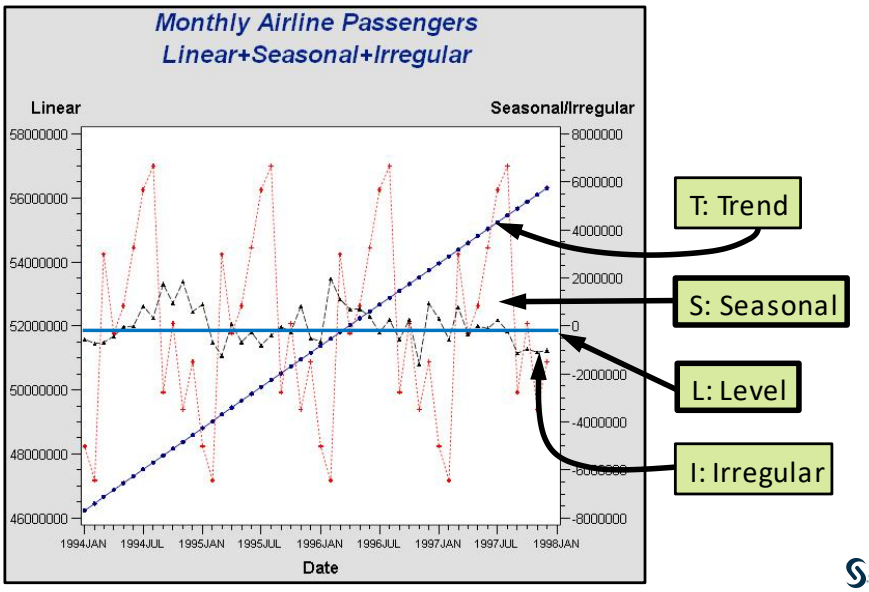
Signal Components: Irregular



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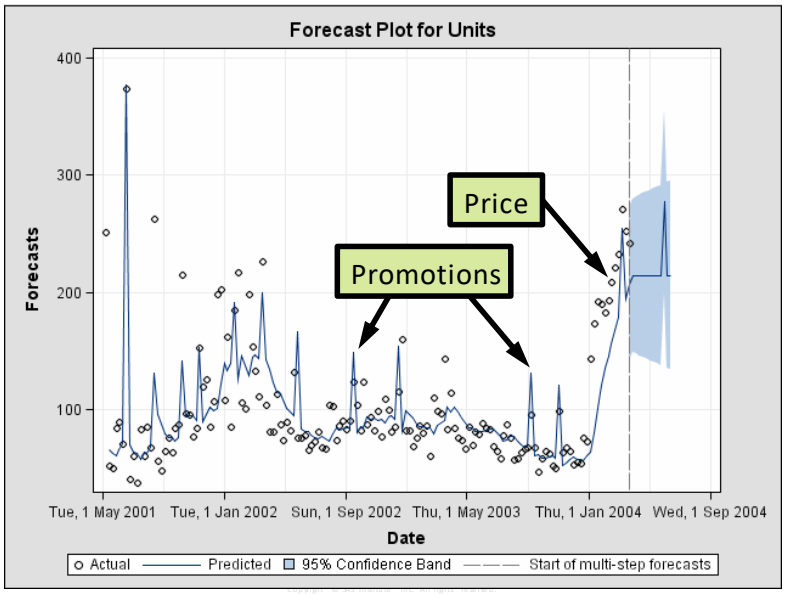
Component Decomposition



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Signal Components: Exogenous Effects (Inputs)



The Retail time series represents weekly unit sales for an item. The series shows evidence of promotion and price effects. Exogenous variables can be related to the target series in complex ways. The exogenous series and the target series can be contemporaneously correlated. (The Y_t is correlated with X_t.) Alternatively, there can be a *dynamic* relationship. (Y_t is related to X_{t-k}.)

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Autocorrelation and White Noise

Autocorrelation

Objectives

- Analyze a time series with respect to signal (systematic variation) and noise (random variation).
- Describe the autocorrelation function plot, white noise test, and other diagnostic tools that are used for assessing a time series.

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Forecasting?





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Forecasting?

The Gambler's Fallacy

- Something that happened more frequently than normal in the past balances out and happens less frequently in the future.
  - Rolling "snake eyes"  two times in a row means that you are less likely to roll it again in the future.
  - Landing on a red-colored number eight times in a row on the roulette wheel means that a black number is more likely on the next spin.



- Can you forecast the next roll of the dice from past rolls in a dice game?

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Correlation of Y with Past Y: *Autocorrelation*

Autocorrelation (Order 1):

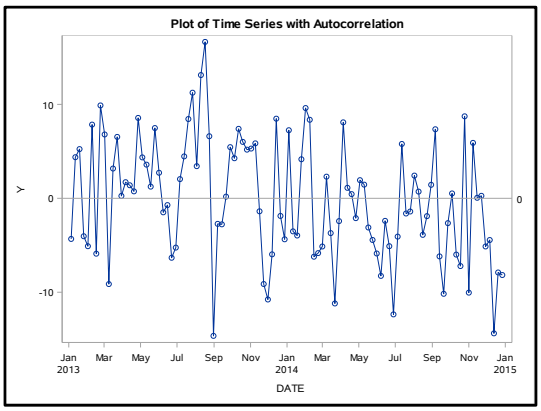
$$Y_t \text{ is correlated with } Y_{t-1}$$

Time Series at Time t:

$$Y_t = Y(t)$$

First Lag:

$$Y_{t-1} = Y(t-1)$$



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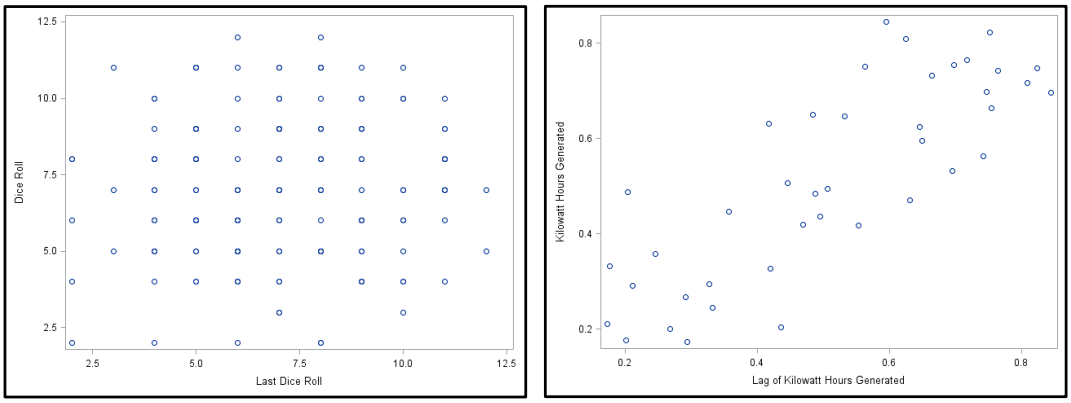
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Autocorrelation simply means that current values in a time series (Y_t) are related with previous values. The correlation between current values and immediately preceding or *lagged* values (Y_{t-1}) is called *first order autocorrelation*. If the correlation extends to the values of two time points that are previous to the current values (Y_{t-2}), that is called *second order autocorrelation*, and so on. Like other correlations, autocorrelations can be either positive or negative with a range between -1 and 1.

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If a series contains trend, seasonality, or both, it contains autocorrelation. Even series that do not contain trend or seasonality can contain autocorrelation. The series on the slide does not appear to contain trend or seasonality (at least visually), but it does contain autocorrelation! When this is the case, detecting autocorrelation visually is, at best, difficult and, at worst, possibly misleading. It is best to focus on specific autocorrelation plots and diagnostics to evaluate the presence of autocorrelation in a time series.

Autocorrelation Scatter Plots



- Autocorrelation is the correlation of present values versus lagged values.
- Autocorrelation between the present value and the first lagged value is called *first order* autocorrelation.

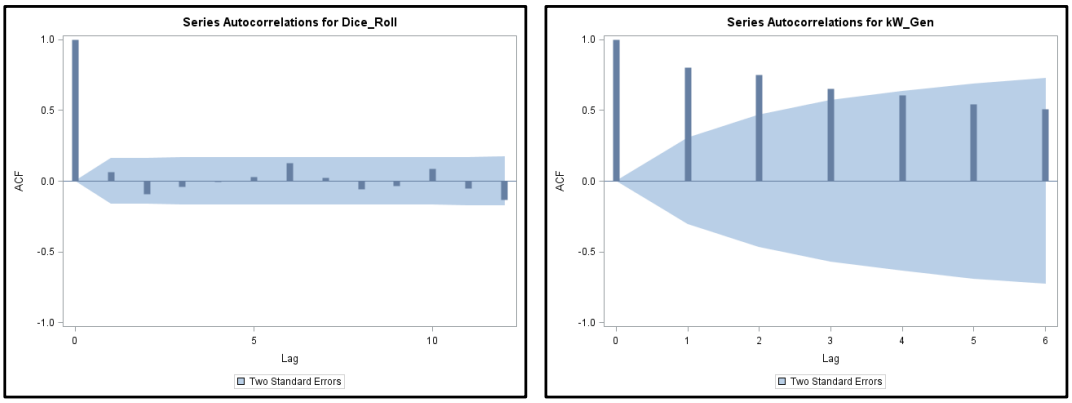
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You could see autocorrelation by creating a column of lagged values in a time series data set and creating a scatter plot. On the left is a plot of the current versus first lagged value of a time series where there is little to no autocorrelation. The right plot shows positive first-order autocorrelation.

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Autocorrelation Plots



- The autocorrelation plot enables you to see the autocorrelation at multiple lags.
- The blue range indicates 95% confidence intervals for each lag.

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Because autocorrelation could theoretically exist with any ordered lag (Y_{t-p}), it is not reasonable to try to create scatter plots for all. Autocorrelation function (ACF) plots are useful as a first step in detecting potential autocorrelation in a time series. Each spike represents the autocorrelation at lag *p*. In addition, 95% confidence interval areas are represented by the blue shaded area.

Where spikes extend beyond the confidence bounds, the autocorrelation is said to be statistically significant at the 0.05 level.

Note: A spike representing the autocorrelation at lag 0 (always equal to 1) is included for comparison.

The left plot shows the ACF plot for the dice roll series. There are no significant spikes. However, there are three significant spikes in the ACF on the right for the kilowatt hours generated series.
It seems that the generated kilowatt hours are a function of past values. (Additional plots are introduced later to further diagnose order.)

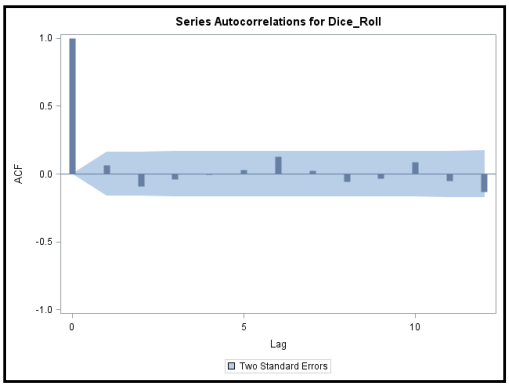
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White Noise

A White Noise Series

A white noise series has the following characteristics:

- varies randomly around its mean
- has no systematic variation
- consists of only random variation
- has constant variance



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Think about the dice roll time series example. You know that the last roll of the dice is not predictive of the next roll. Dice rolls are governed by a random process. The expected average number of dots shown in a roll of two dice is 7, and the expected standard deviation is 2.41. It is not important whether the dice roll is the first, the seventh, or the 670,000th. The mean and variance should remain constant. Each dice roll is independent of all other dice rolls. In time series terminology, this is considered a *white noise* series.

A white noise series, by definition, has no autocorrelation. If you want to forecast the next value of a white noise series, your best guess is always the mean of the series.

Is white noise in a time series good or bad?

If the series is white noise, it means that your best forecasts are simply the mean of the series, that is, a forecast with high variance. However, if the residual values (actual minus predicted values) from the forecasting model are white noise, it indicates that the elements that you included in your model adequately explained all that is explainable (the signal) in the series. Therefore, that which was not explained by the model is not explainable.

A white noise series technically implies a mean of 0, although even a series with a nonzero mean can be considered white noise, if the series of deviations from the mean ($Y_t - \bar{Y}_t$) are white noise.

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The Ljung-Box Chi-Square Test for White Noise

- A *white noise* time series is a Gaussian (normal, bell-shaped) time series with a mean of zero and a positive fixed variance in which all observations are independent of each other.
- The null hypothesis is that the series is white noise, and the alternative hypothesis is that one or more autocorrelations up to lag *m* are not zero.

  H_0: The series is white noise.

  H_1: The series is ***not*** white noise.

Note: The Ljung-Box test can be applied to the original series or to the residuals after you fit a model.

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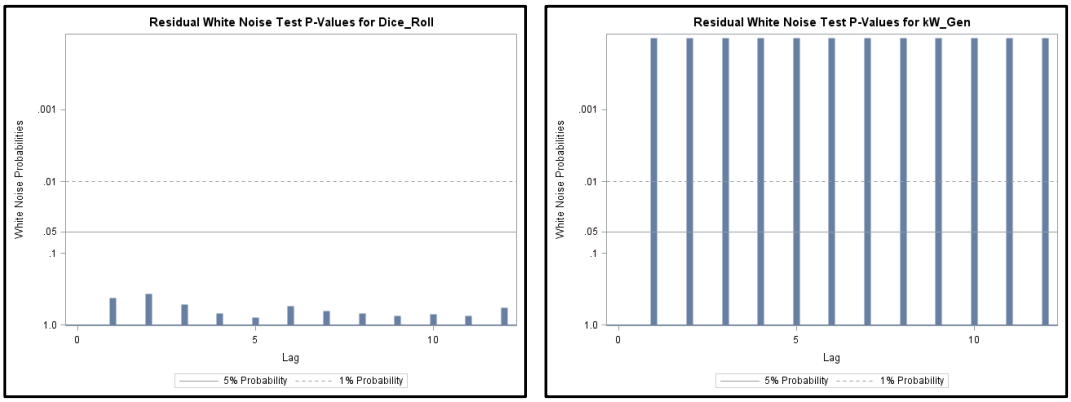
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A popular test for white noise is the Ljung-Box test. The test statistic is calculated as follows:

$$\chi^2_m = n(n+2)\sum_{k=1}^{m}\frac{r_k^2}{n-k},\ r_k = \mathrm{ACF}(k), \text{ given } \mu = 0.$$

The statistic is *cumulative*. This means that the null hypothesis is that all of the autocorrelations, up to and including lag *m*, are white noise. A rejection of the null hypothesis at level *m* does not inform which lags cause the significant result.

The Ljung-Box Chi-Square Test for White Noise



"White means white."

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The plot of the Ljung-Box chi-square test can be used to quickly assess whether the autocorrelation at any lag rejects the white noise assumption.

Note: Notice the scale and ordering of values on the Y axis. The order is descending from bottom to top, and the probability values (representing *p*-values) are not linearly scaled. This representation enables you to see statistical significance at various significance levels (0.10, 0.05, and 0.01) more easily. It also means that non-significant tests (high *p*-values) are represented by short spikes. Hence, the expression "White means white" when you glance at the plot.

The white noise plot is sometimes displayed with the Y-axis values in ascending order from bottom to top. In that representation, long bars represent high *p*-values, and therefore, white noise. You should pay attention to the Y-axis values before you draw conclusions about the white noise tests.

The Ljung-Box Chi-Square Test for White Noise

White Noise Series

Autocorrelation Check for White Noise									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	5.00	6	0.5436	0.065	-0.095	-0.038	-0.004	0.028	0.129
12	10.39	12	0.5821	0.026	-0.060	-0.033	0.087	-0.054	-0.131

Autocorrelated Series

Autocorrelation Check for White Noise									
To Lag	Chi-Square	DF	Pr > ChiSq	Autocorrelations					
6	121.35	6	<.0001	0.804	0.750	0.652	0.608	0.546	0.509
12	147.33	12	<.0001	0.477	0.384	0.265	0.158	0.095	0.027

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These tables of white-noise tests show cumulative results for six lags at a time. The top table shows non-significance up to lag 6 and also to lag 12. No individual autocorrelation value exceeded 0.131 in absolute value. This came from the dice roll data. The bottom table shows the white noise tests for a series that has autocorrelation. Notice that even though the test for white noise to lag 12 is statistically significant, none of the last three autocorrelations was greater than 0.158. Remember that these tests are cumulative.

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Forecasting and Accuracy Assessment

Objectives

- Describe the honest assessment process from start to finish.
- Define a fit sample and a holdout sample in time series forecasting.
- Describe common error measures that can be used to evaluate forecast accuracy.

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Forecasting

If someone asks you whether you can forecast something, your answer should always be "Yes."

If someone asks you whether you can forecast something ***accurately***, you cannot answer until you establish what accuracy means and until you perform preliminary diagnostics of the series.

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Liability

"Do you stake your reputation on the accuracy of these forecasts?"

"No, but I stake my reputation on the methodology that was used to generate the forecasts."

- You might have no control over data accuracy and validity.

For example, is modeling the volume sold an accurate representation of true demand?

Were there supply shortages that you were unaware of that could hurt forecasts?

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Liability

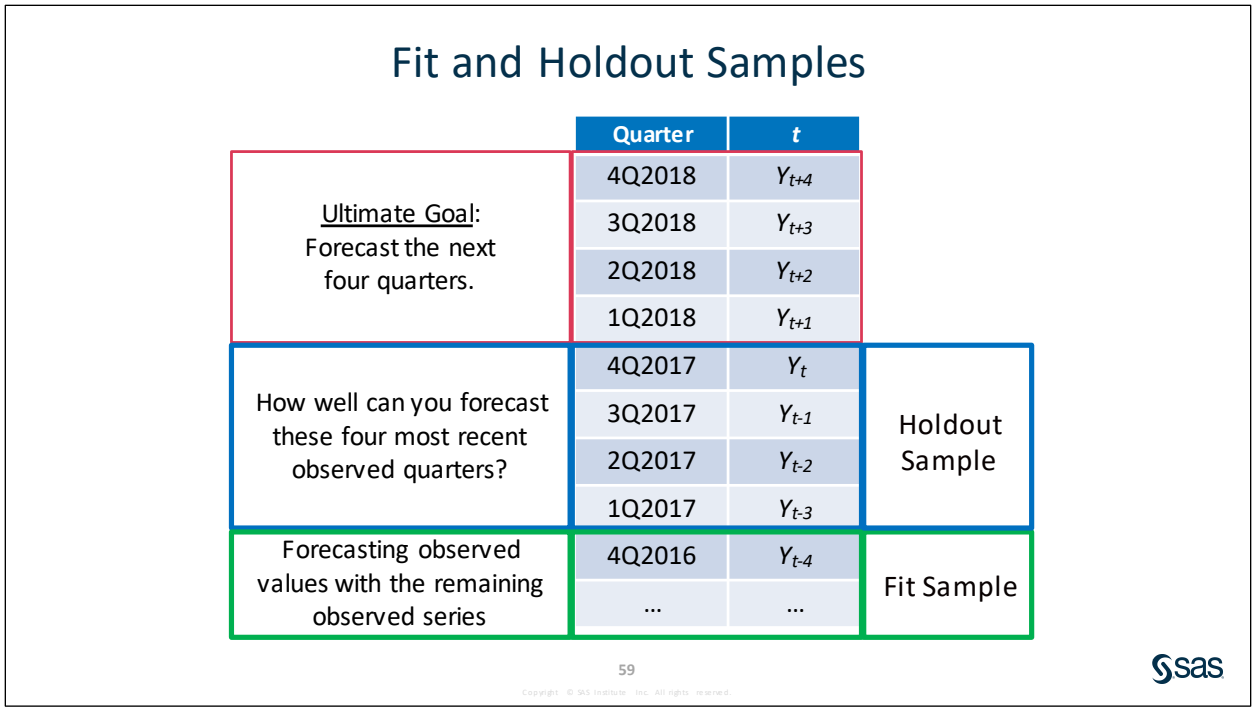
- You need to assume that the underlying future behavior remains consistent with past behavior.
- However, you have no control over future events that might affect future behavior, such as catastrophes, economic downturns, war, the integrity of key players, the survival of key players, and so on.

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Time series analysis, like other branches in statistics, can be grouped into two broad segments:

- inference-based analysis
- prediction analysis

For those who are intent on forecasting future, unobserved periods, it is a best practice to split the data set into a fit sample and a holdout sample. However, unlike other forms of predictive modeling, where the holdout sample is a random or stratified sample from the original data, the holdout sample in time series forecasting is the final k values of the series.

The fit sample is used to derive the forecast model. The holdout sample is used to evaluate how well the forecast model predicts observations that were not used to build the model. Simply put, if your objective is to forecast the next four quarters of revenue, you want to build a model that accurately forecasts the most recently observed four quarters. (The slides below provide rules of thumb for selecting the holdout sample.)

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Honest Assessment: Simulating a Prospective Study

1. Divide the time series data into two segments.
   - The *fit sample* is used to derive a forecast model.
   - The *holdout sample* is used to evaluate forecast accuracy.
2. Derive a set of candidate models.
   - Exponential Smoothing Models – Chapter 2
   - ARMA and ARMAX Models – Chapter 3
   - Unobserved Components Models – Chapter 4
3. Calculate the chosen model accuracy statistic for each model by using the fit sample to forecast the holdout sample.
4. Choose the model with the best accuracy statistic.
5. Using the best model, generate forecasts for *n* future periods.

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Choosing the Holdout Sample

- Choose enough time points to cover a complete seasonal period. For example, for monthly data, hold out at least 12 observations.
- The holdout sample is always at the end of the series.
- If unique behavior occurs within the holdout sample, do not use a holdout sample. Instead, base accuracy calculations on the entire series.
- If there is insufficient data to split into a fit and holdout samples, base accuracy calculations on the entire series.

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Summary of Data Used for Forecast Model Building

Fit Sample	Holdout Sample
• **Used to estimate model parameters for accuracy evaluation** • **Used to forecast values in holdout sample**	• **Used to evaluate model accuracy** • **Simulates retrospective study**



Full = Fit + Holdout data is used to fit a deployment model.

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Rules of Thumb

- At least four time points are required for every parameter to be estimated in a model.
- Anything above the minimum series length can be used to create a holdout sample.
- Holdout samples should rarely contain more than 25% of the series.

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continued…

Model Fit Statistics

Mean Absolute Percent Error:

$$\text{MAPE} = \left(\frac{1}{n} \sum_{t=1}^{n} | Y_t - \hat{Y}_t | / Y_t \right) * 100\%$$

Mean Absolute Error:

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^{n} | Y_t - \hat{Y}_t |$$

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Both MAPE and MAE are very common measurements of model fit. When you choose between candidate models, the model with the ***lowest*** value for MAPE or for MAE is the model that fits the holdout data the best.

Notice that both statistics take the absolute value of the observed values minus the predicted values. Mathematically, this is done by necessity, but it might omit important information about the fit of the model.

For example, one model might have a very low MAPE or MAE, but it constantly under fits. (That is, the predicted values always fall short of the observed values in the holdout data set.) Looking at MAPE or MAE alone does not always paint the whole picture. Instead, look at the value of MAPE in conjunction with a plot of observed and predicted values in the holdout sample to assess the model fit.

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Model Fit Statistics

R-Square: $R^2 = 1 - \sum_{t=1}^{n}(Y_t - \hat{Y}_t)^2 \Big/ \sum_{t=1}^{n}(Y_t - \bar{Y})^2$

Root Mean Square Error:

$$\text{MSE}^* = \frac{1}{n-k}\sum_{t=1}^{n}(Y_t - \hat{Y}_t)^2$$

$$\text{RMSE} = \sqrt{\text{MSE}}$$

* For holdout samples, use divisor *n* rather than *n-k*.

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Simple Example for MAPE and MAE

Holdout Sample	Y observed	$\hat{Y}$ forecasted	$Y - \hat{Y}$ residual	$\lvert Y - \hat{Y} \rvert$	$\lvert Y - \hat{Y} \rvert / Y$
Y_t	105	109	-4	4	0.038
Y_{t-1}	120	119	1	1	0.008
Y_{t-2}	128	125	3	3	0.023
Y_{t-3}	122	120	2	2	0.016

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|Y_t - \hat{Y}_t| \qquad \text{MAPE} = \left(\frac{1}{n}\sum_{t=1}^{n}|Y_t - \hat{Y}_t| / Y_t\right) * 100\%$$

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Simple Example for MAPE and MAE

Holdout Sample	Y observed	$\hat{Y}$ forecasted	$Y - \hat{Y}$ residual	$\lvert Y - \hat{Y} \rvert$	$\lvert Y - \hat{Y} \rvert / Y$
Y_t	105	109	-4	4	0.038
Y_{t-1}	120	119	1	1	0.008
Y_{t-2}	128	125	3	3	0.023
Y_{t-3}	122	120	2	2	0.016

$$MAE = \frac{4+1+3+2}{4} \qquad MAPE = \left(\frac{.038+.008+.023+.016}{4} \right) * 100\%$$

$MAE = 2.5$ $MAPE = 2.1\%$

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Necessary Conditions for Good Forecasts

- The identified signal continues into the future.
- Forecasting model complexity should be adequate to capture signal components.
- Forecasting models should not be overly complex.
- The best forecasting model is the one that captures and extrapolates the most signal, and that also ignores the noise.

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Choosing a Winning Set of Forecasts

Good forecasts should have the following characteristics:

- be highly correlated with the actual series values
- exhibit small forecast errors
- capture the prominent features of the original time series

In addition, assessment of forecast quality should be based on the business, engineering, or scientific problem that is being addressed.

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Time Series Models

Objectives

- List the three families of time series models that are illustrated in this course.
- Describe the main features of each model family.
- Give examples of using an appropriate model from each family to fit the **DOTAIR9497** data.
- Discuss which model family is preferred for different types of time series analyses.

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Necessary Conditions for Good Forecasts

- The identified signal continues into the future.
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Time Series Models in This Course

- exponential smoothing (ESM)
- autoregressive moving average [with exogenous variables] (ARMA[X])
- unobserved components (UCM)

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Exponential Smoothing Models

Exponential Smoothing Premise

- Weighted averages of past values can produce good forecasts of the future.
- The weights should emphasize the most recent data.
- Forecasting should require only a few parameters.
- Forecast equations should be simple and easy to implement.

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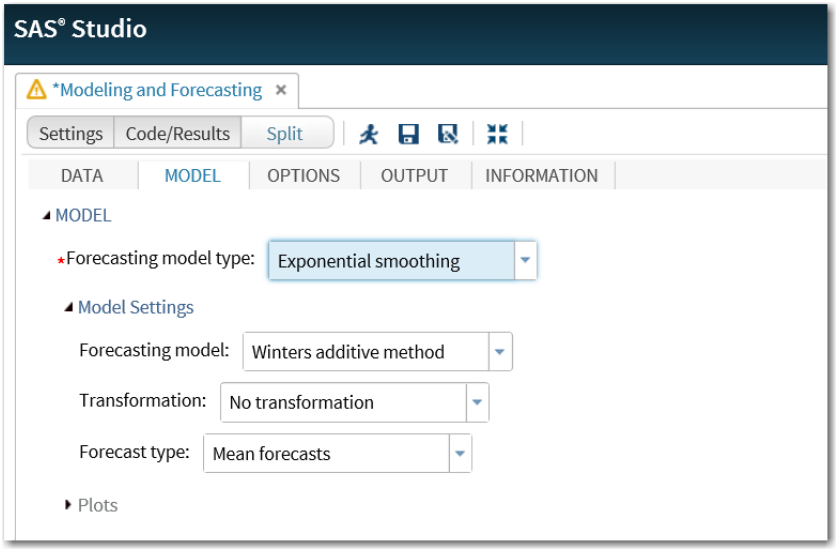
ESM Parameters and Keywords

ESM	Parameters	Option Name
Simple	ω	SIMPLE
Double	ω	DOUBLE
Linear (Holt)	ω, γ	LINEAR
Damped-Trend	ω, γ, ϕ	DAMPTREND
Seasonal	ω, δ	SEASONAL
Additive Winters	ω, γ, δ	ADDWINTERS
Multiplicative Winters	ω, γ, δ	WINTERS

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Exponential Smoothing Model Implementation



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Two of the seven exponential smoothing models accommodate trend, seasonal, and irregular variation (Winters additive and Winters multiplicative). The Modeling and Forecasting task in SAS Studio is used above to specify a Winters additive method on the **DOTAIR9497** data.

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Exponential Smoothing Model Implementation

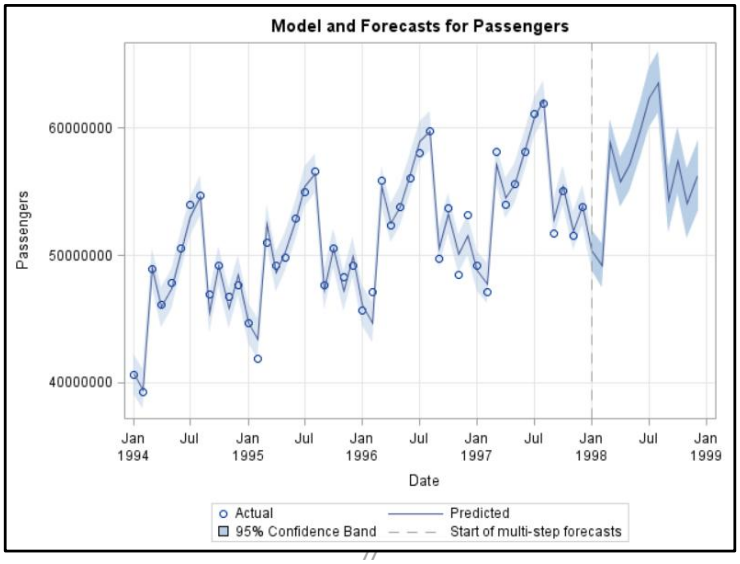
```
proc sort data=STSM.DOTAIR9497 out=WORK.TempSorted;
        by DATE;
run;

/*****************************************************************************************/
/* Exponential Smoothing                                                                 */
proc esm data=WORK.TempSorted back=0 lead=12 seasonality=12 plot=(corr errors
                 modelforecasts);
        /* id statement */
        id DATE interval=month;
        forecast Passengers / alpha=0.05 model=addwinters transform=none;
run;
```

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The ESM procedure (SAS/ETS) syntax is generated by SAS Studio. It fits the Winters additive (ADDWINTERS) specification and forecasts 12 months into the future.

Exponential Smoothing Model Forecasts



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The selected ESM model seems to do a good job of capturing and extrapolating the trend and seasonal components of the data.

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ARIMAX Models

ARMAX Models

ARMAX: **A**uto**R**egressive **M**oving **A**verage with e**X**ogenous variables

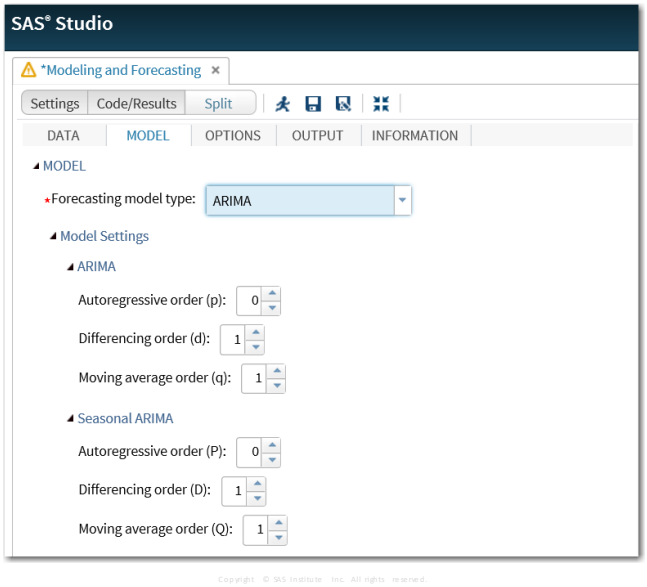
- AR: Autoregressive ⇨ Time series is a function of its own past.
- MA: Moving Average ⇨ Time series is a function of past shocks (deviations, innovations, errors, and so on).
- X: Exogenous ⇨ Time series is influenced by external factors.

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Integration (the “I” in ARIMA) corresponds to the accommodation of nonstationary variation in an ARIMA model. Trend and seasonal components are examples of nonstationary variation. This is because, if a series has trend or seasonal components, its mean is a function of time. The mean of a stationary series is well defined. It is not a function of time. (A later chapter provides examples of stationary series.)

Box-Jenkins ARIMAX Implementation



An appropriate ARIMA model for the **DOTAIR9497** data set is the classic Box-Jenkins Airline model for series G. (It is shown above.) This model accommodates trend and seasonality with a first and seasonal (12) span differences. Differencing orders are offset by moving average orders at the same lags.

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Box-Jenkins ARIMAX Implementation

```
proc sort data=STSM.DOTAIR9497 out=WORK.TempSorted;
        by DATE;
run;

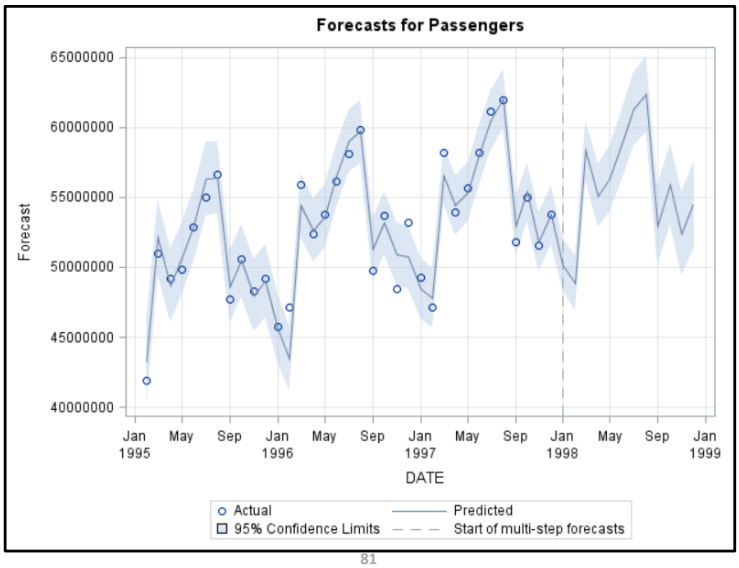
/*****************************************************************************************/
/* ARIMA or ARIMAX                                                                       */
proc arima data=WORK.TempSorted plots
        (only)=(series(corr crosscorr) residual(corr normal) forecast(forecastonly));
        identify var=Passengers(1 12);
        estimate q=(1) (12) method=ML;
        forecast lead=12 back=0 alpha=0.05 id=DATE interval=month;
        outlier;
quit;
```

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In the generated, ARIMA procedure syntax, differencing orders are specified in the IDENTIFY statement. The Q option in the ESTIMATE statement specifies moving average orders. (More details about the ARIMA procedure syntax are provided later.)

Box-Jenkins ARIMAX Forecasts



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The ARIMA specification fitted above does a good job of capturing and extrapolating the trend, seasonal, and irregular components in the Passengers series.

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Unobserved Components Models

Unobserved Components Models (UCMs)

- UCMs are also known as *structural time series models*.
- Decompose time series into components:
  - trend
  - season
  - cycle
  - irregular
  - regressors
- General form:

$$Y_t = \text{Trend} + \text{Season} + \text{Cycle} + \text{Regressors}$$

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UCMs

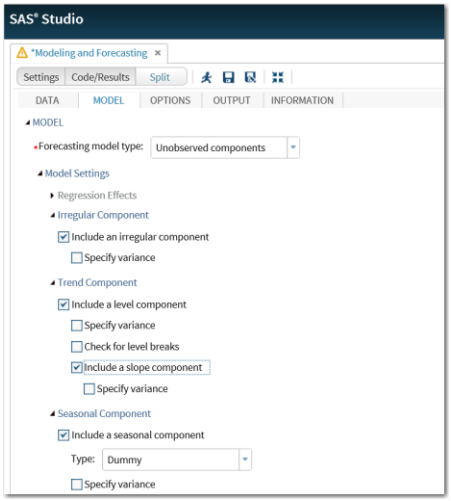
- Each component captures some important feature of the series dynamics.
- Components in the model have their own models.
- Each component has its own source of error.
- The coefficients for trend, season, and cycle are dynamic.
- The coefficients are testable.
- Each component has its own forecasts.

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UCM Implementation



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A UCM model that accommodates the systematic components in the **DOTAIR9497** data set is shown above. To create an appropriate UCM, a statement that corresponds to each component of hypothesized, systematic variation is specified.

Note: The LEVEL and SLOPE components (statements) combine to fit the trend in the **DOTAIR9497** data.

UCM Implementation

```
proc sort data=STSM.DOTAIR9497 out=WORK.TempSorted;
        by DATE;
run;

/*****************************************************************************************/
/* Unobserved Components                                                                 */
proc ucm data=WORK.TempSorted;
        id DATE interval=month;
        model Passengers;
        irregular;
        level;
        slope;
        season length=12 type=dummy;
        forecast lead=12 back=0 alpha=0.05;
        outlier;
run;
```

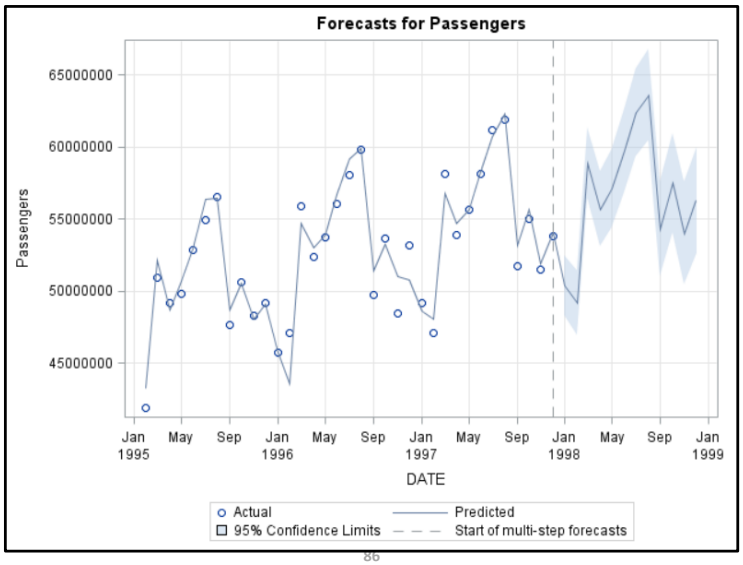
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A statement for each of the components of variation in the **DOTAIR9497** data set is seen in the generated syntax above.

UCM Forecasts



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The UCM specification does a good job of capturing and extrapolating the systematic variation in the **DOTAIR9497** data.

Choosing the Right Model for the Job

Usability

Best to worst:

1. ESM
2. UCM
3. ARMAX

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Complexity

Least to Most:

1. ESM
2. ARMAX
3. UCM

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Robustness

Best to worst:

1. ESM
2. ARMAX
3. UCM

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Ability to Accommodate Dynamic Regression Effects

Best to worst:

1. ARMAX
2. UCM
3. ESM

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Idea Exchange

List the types of forecasting models that you used in your analyses.

Give an example of an analysis that would be well suited to each of the following families of models:

- ESM
- ARIMAX
- UCM

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Quick Demonstrations

Time Series in SAS Studio on SAS9

Time Series Creation

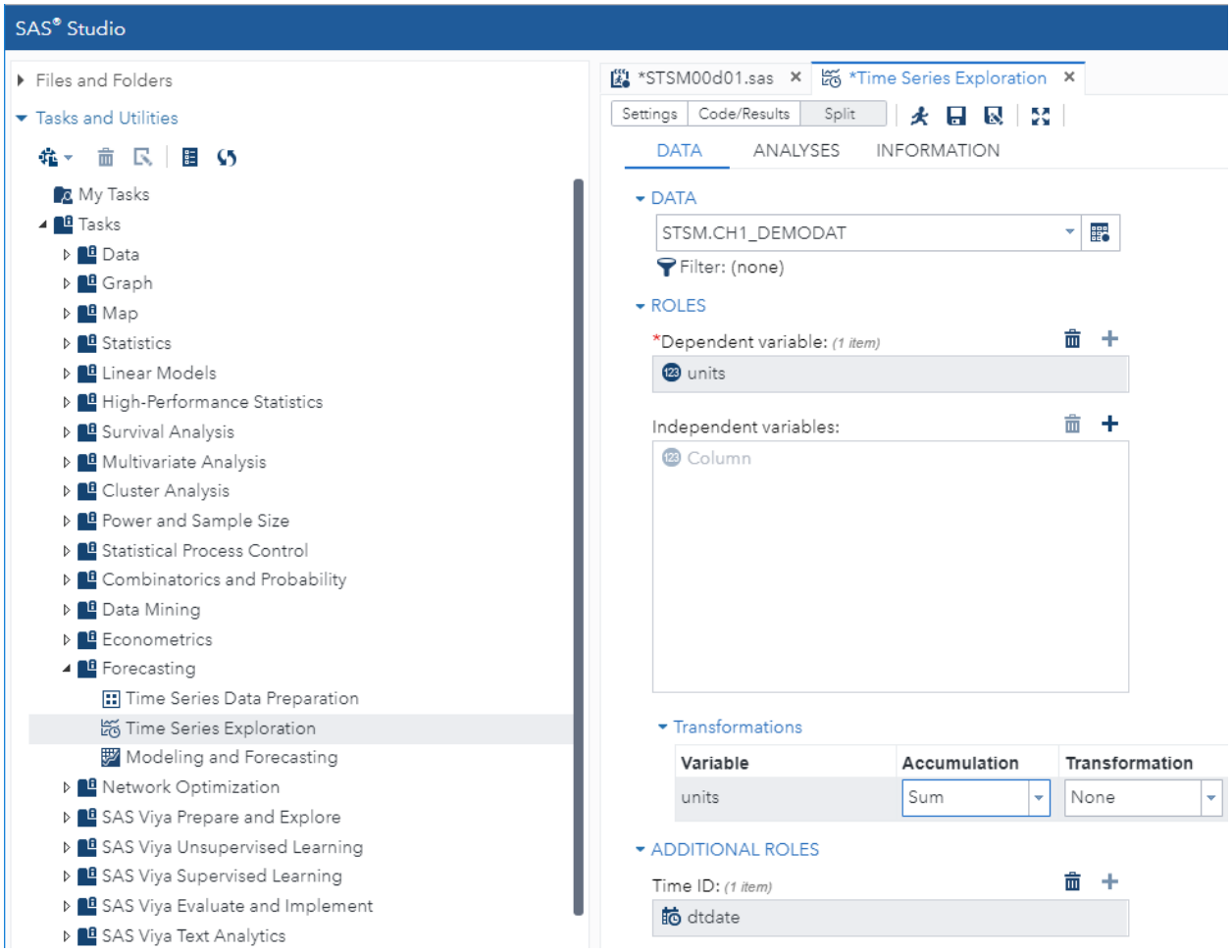
This demonstration provides examples of creating time series through the process of accumulation. The Time Series Exploration task in SAS Studio is used to illustrate this. The transactional data set for the analysis is **STSM.CH1_DEMODAT**. The dependent variable is **units** of sales. There are two timestamps: **date** is a date variable and **dtdate** is a datetime variable. A portion of the data is shown below. The transactional observations begin on 02January2004 and end on 29December2006 inclusive.

Obs	units	dtdate	date	mon	yr
1	.	02JAN2004:08:00:00	02JAN2004	1	2004
2	1940	02JAN2004:09:00:00	02JAN2004	1	2004
3	.	02JAN2004:10:00:00	02JAN2004	1	2004
4	3147	02JAN2004:11:00:00	02JAN2004	1	2004
5	.	02JAN2004:12:00:00	02JAN2004	1	2004
6	.	02JAN2004:13:00:00	02JAN2004	1	2004
7	.	02JAN2004:14:00:00	02JAN2004	1	2004
8	.	02JAN2004:15:00:00	02JAN2004	1	2004
9	.	02JAN2004:16:00:00	02JAN2004	1	2004
10	.	02JAN2004:17:00:00	02JAN2004	1	2004

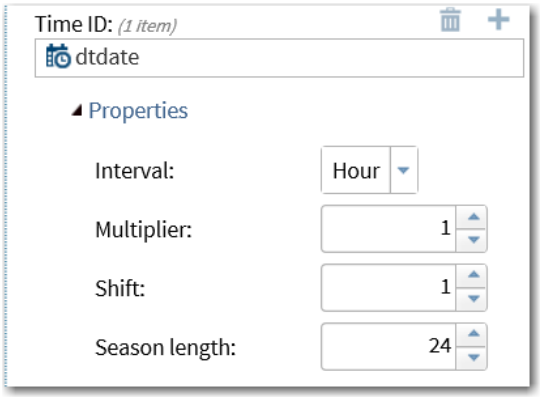
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Examining the Data on an Hourly Interval

1. Assign **units** as the dependent variable. Expand the **ADDITIONAL ROLES** property, and assign the **dtdate** column as the time ID.
2. Select **Sum** for the **Accumulation** field.



3. After the time ID variable is set, a one-hour interval that begins in the first hour of a 24-hour day is detected as the natural interval of the data. (The options shown below can be changed to modify the detected interval.)



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Note: The TIMEID procedure in SAS/ETS runs “under the hood” in SAS Studio to detect the interval of the data.

4. Click  on the toolbar to maximize the view.
   The TIMESERIES procedure in SAS/ETS provides the main functionality for the Time Series Exploration task. A subset of the contents of the CODE window is shown below.

```
proc sort data=STSM.CH1_DEMODAT out=Work.preProcessedData;
   by dtdate;
run;

proc timeseries data=Work.preProcessedData seasonality=24
                plots=(series corr);
   id dtdate interval=hour;
   var units / accumulate=total transform=none dif=0 sdif=0;
run;
```

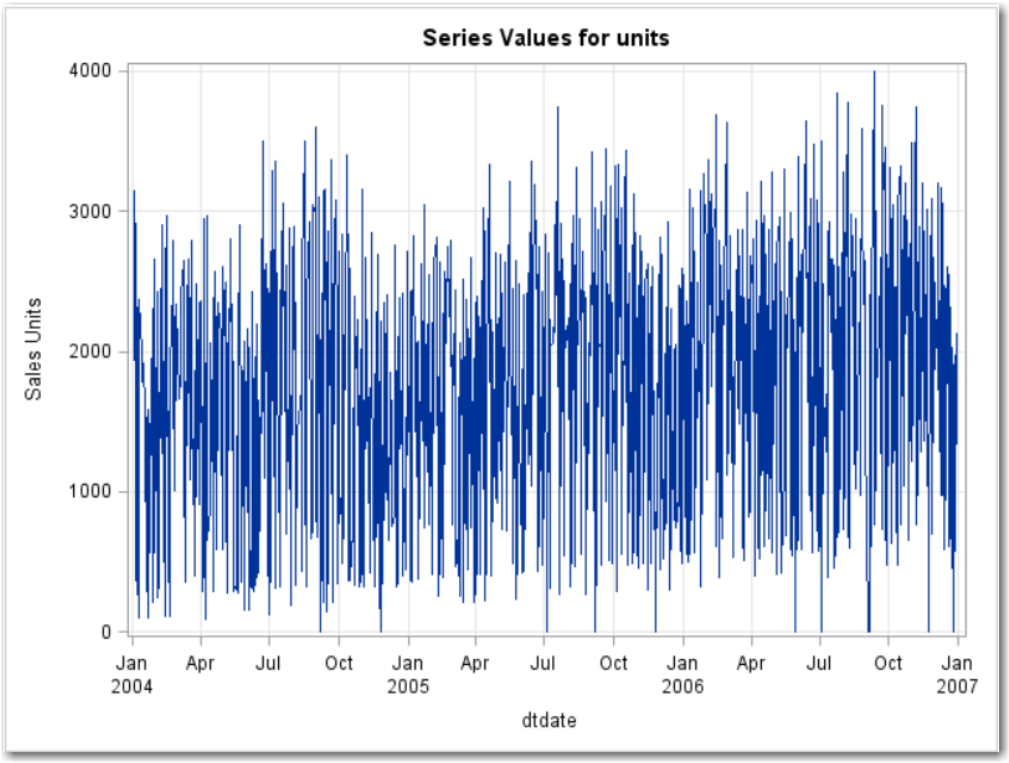
- The SORT procedure is used to create a working copy of the transactional data and sort it from the earliest to the latest date. In different versions of SAS Studio, the naming convention of the sorted data might be different.
- The TIMESERIES procedure is used to accumulate the transactional data to an hourly interval. The PLOTS option in the procedure statement creates the series plot and correlation panel. The ID statement defines the datetime ID variable and the interval and accumulation options that correspond to the settings that are selected on the DATA tab.

Note: If you produce the code by entering it directly in the editor, the code below produces the necessary output. This code assumes that the data in **stsm.ch1_demodat** are properly sorted.

```
proc timeseries data=stsm.ch1_demodat out=out_totalhour
                plots=(series);
   id dtdate interval=dthour accumulate=total;
   var units;
run;
```

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5. To submit the code and run the Time Series Exploration task, click **Run** on the toolbar. The series plot is shown below. There are many hourly intervals in a three-year time span.



6. To get a better look at the hourly, intervaled time series, the data is truncated and then re-plotted. Because there is currently no option to do this with the DATA properties of SAS Studio, the PROC TIMESERIES syntax is edited directly. Click the **CODE** tab and then click **Edit**.



Note: The syntax is now contained on a new tab.

7. Add an END option to the ID statement in PROC TIMESERIES as shown below. The end value *12jan2004:00:00* truncates the data so that only the first 10 days of data are shown in the plots.

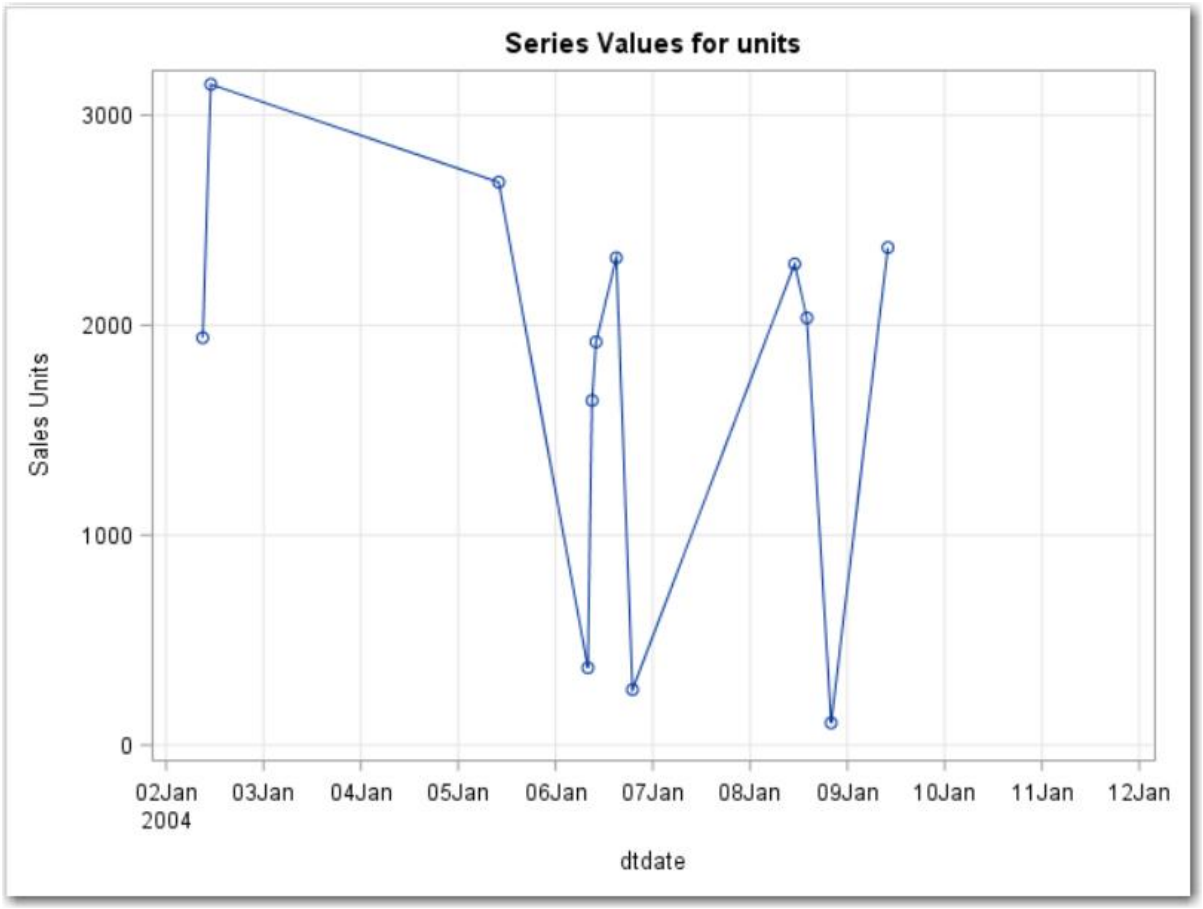
```
proc timeseries data=Work.preProcessedData seasonality=24
                plots=(series corr);
   id dtdate interval=hour end='12jan2004:00:00'dt;
   var units / accumulate=total transform=none dif=0 sdif=0;
run;
```

Note: If you produce the code by entering it directly in the editor, the following code produces the necessary output:

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```
proc timeseries data=out_totalhour plots=(series);
   id dtdate interval=dthour accumulate=total
              end='12jan2004:00:00'dt;
   var units;
run;
```

8. Click **Run** to submit the modified syntax.



One important point, illustrated in the plot above and the table below, is that issues with missing observations cannot be discerned until after the data is accumulated. If an hourly or daily interval is used for forecasting, a data imputation method might be necessary.

9. Click the **OUTPUT DATA** tab to view the hourly, intervaled, time series data.

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	dtdate	units
1	02JAN04:08	.
2	02JAN04:09	1940
3	02JAN04:10	.
4	02JAN04:11	3147
5	02JAN04:12	.
6	02JAN04:13	.
7	02JAN04:14	.
8	02JAN04:15	.
9	02JAN04:16	.
10	02JAN04:17	.
11	02JAN04:18	.

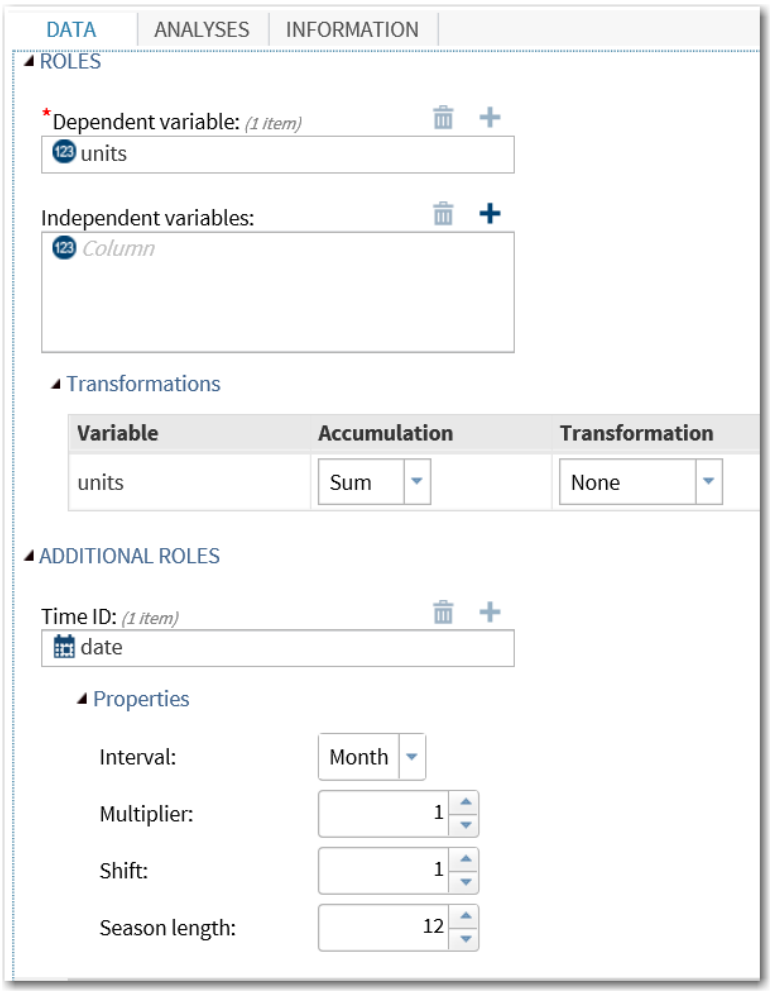
Note: The SETMISSING option in the ID statement of the TIMESERIES procedure can be used to replace missing values in time series with imputed values.

Note: A preferred method for missing value imputation in time series is replacing missing values with their one-step-ahead forecast. The ESM procedure in SAS/ETS can be used for this. See the REPLACEMISSING option in the FORECAST statement.

Creating the Time Series on a Monthly Interval

1. Click the **Time Series Exploration 1** tab to return to the original analysis.
2. Remove the assignment of the time ID variable. Highlight **dtdate** and click the **Trash Can** icon.
3. Assign **date** as the time ID variable and change the interval property to **Month**.
4. Expand the **Transformations** properties and set the **Accumulation** field to **Sum**.
   These steps are summarized in the display below.

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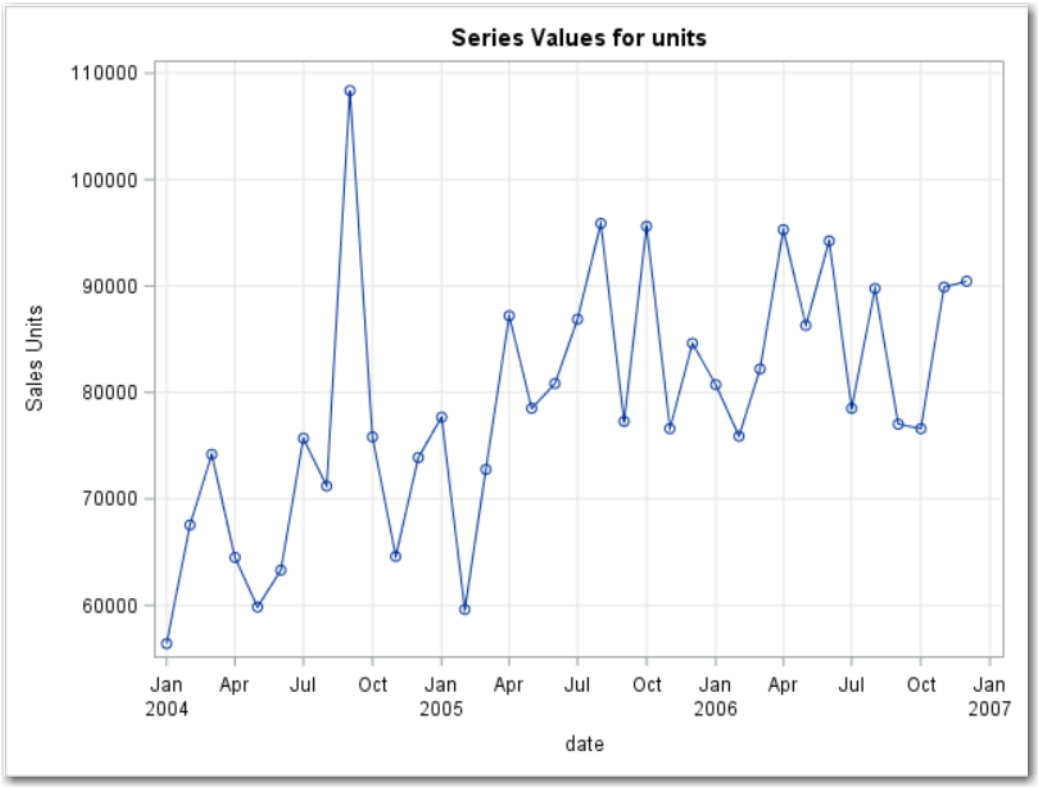


Note: If you produce the code by entering it directly in the editor, the following code produces the necessary output:

```
proc timeseries data=stsm.ch1_demodat out=out_totalmonth
                plots=(series);
   id date interval=month accumulate=total;
   var units;
run;
```

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10. Click **Run** to submit the modified Time Series Exploration task. A plot of the monthly, intervaled **units** time series is shown below.



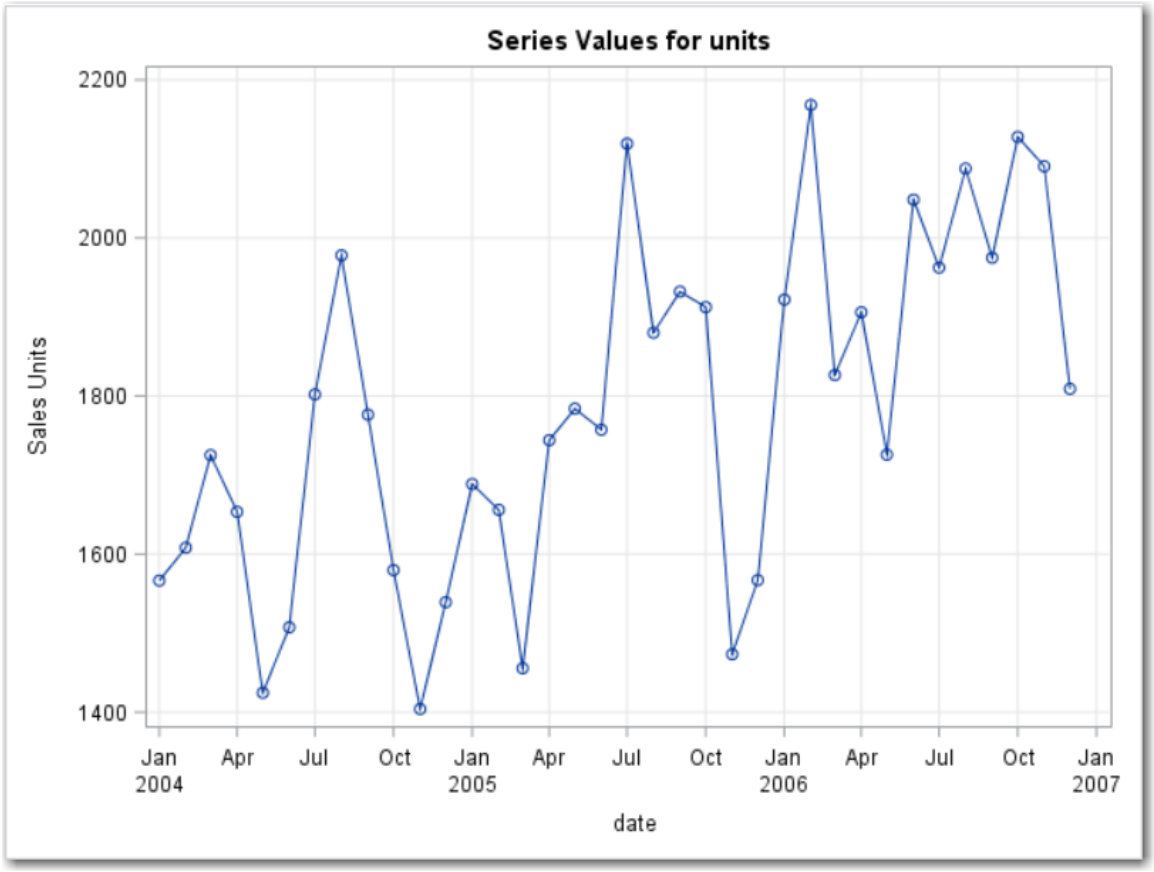
Using a Total or Sum accumulation method, the monthly, intervaled data shows what might be a structural break near the middle of 2005. The spike in Sep2004 might be an anomaly that requires further investigation and consideration when you build a model to accommodate the variation in this series.

11. Change the **Accumulation** field value to **Average**.
12. If you produce the code by entering it directly in the editor, the following code produces the necessary output:

```
proc timeseries data=stsm.ch1_demodat out=out_totalmonth
                plots=(series);
   id date interval=month accumulate=average;
   var units;
run;
```

13. Click **Run**.

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The plot of the time series that was created using an Average accumulation method reveals a trend and a cycle in the time series data.

End of Demonstration

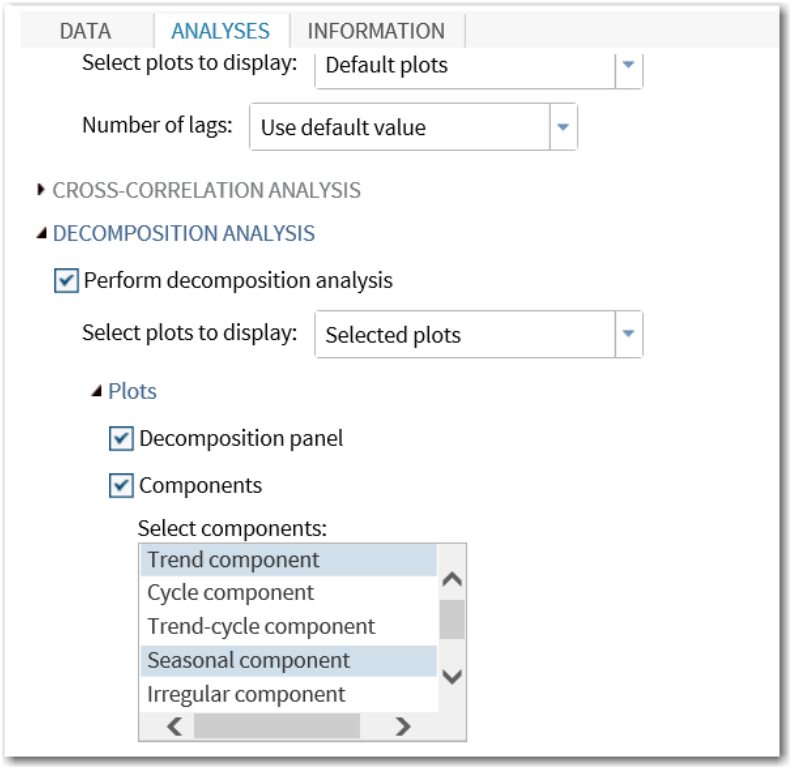
Time Series Identification

This demonstration begins where the previous demonstration ended. The monthly, intervaled time series that was accumulated using the Average method is decomposed to assess the presence of trend, seasonal, and other components of systematic variation.

1. Click the **ANALYSES** tab in the Time Series Exploration task that you created in the previous demonstration.
2. Expand the **DECOMPOSITION ANALYSIS** properties.
3. Select the **Perform decomposition analysis** check box and change the **Select plots to display** field to **Selected plots**.
4. Select the **Components** check box and highlight the **Trend** and **Seasonal** component plots.

These steps are summarized below.

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5. Use SAS/ETS program code.

   Additional syntax includes the following decomposition plot options:

   - TC = Trend Component
   - SC = Seasonal Component
   - IC = Irregular Component
   - CORR plots correlation functions

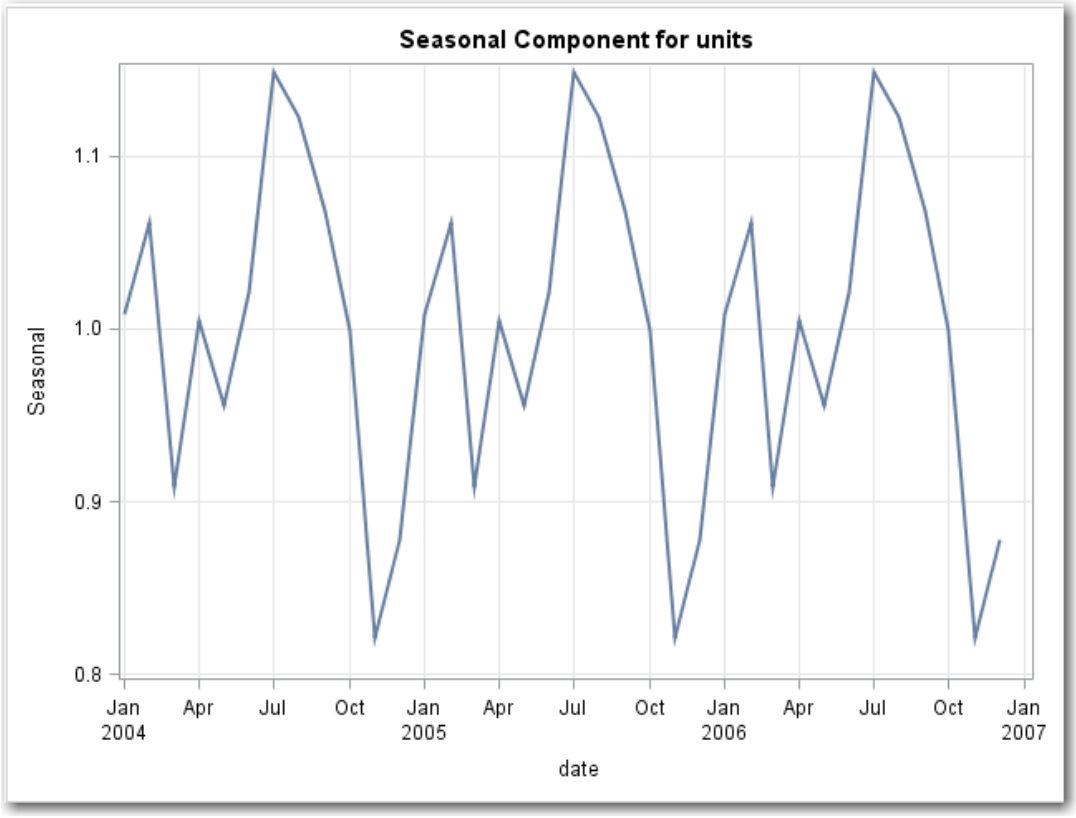
   The **decomp_total** data set contains the decomposition statistics and the *de-seasonalized series*. (The estimated seasonal component is subtracted from the original series to create the output series.)

```
proc timeseries data=stsm.ch1_demodat out=out_totalmonth
                outdecomp=decomp_total
                print=(descstats seasons decomp)
                plot=(series corr tc sc ic);
   id date interval=month accumulate=average;
   var units;
run;
```

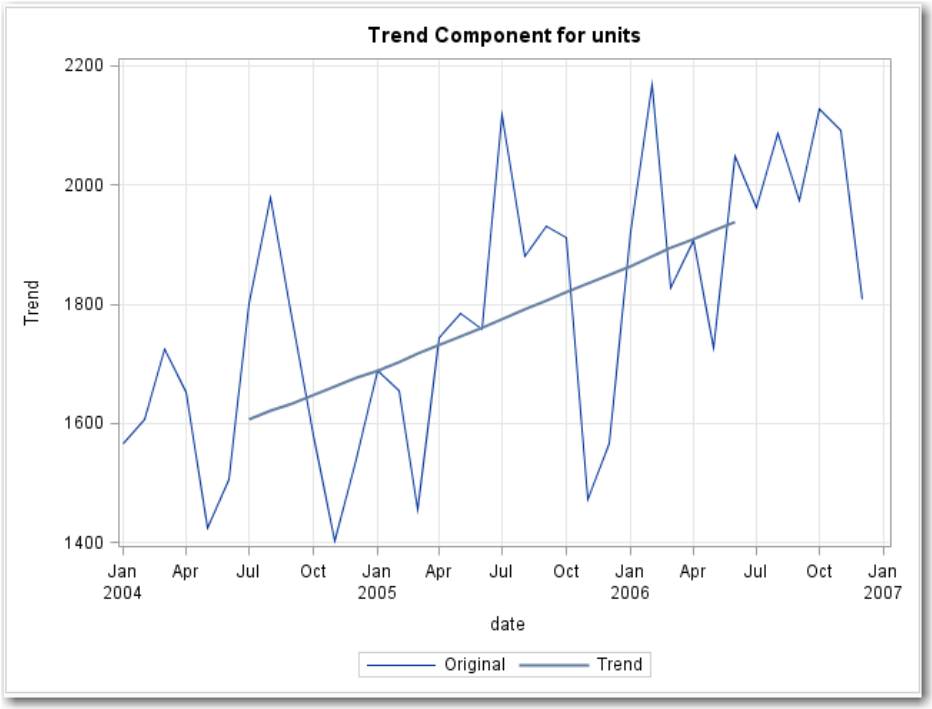
6. Click **Run**.

   The seasonality plot reveals a strong seasonal cycle in the data that was accumulated to a monthly average. The seasonal peak month is July. Sales of **units** in July average more than 10% above average, annual sales. November is the seasonal low month. Sales in November average are almost 20% below the annual average.

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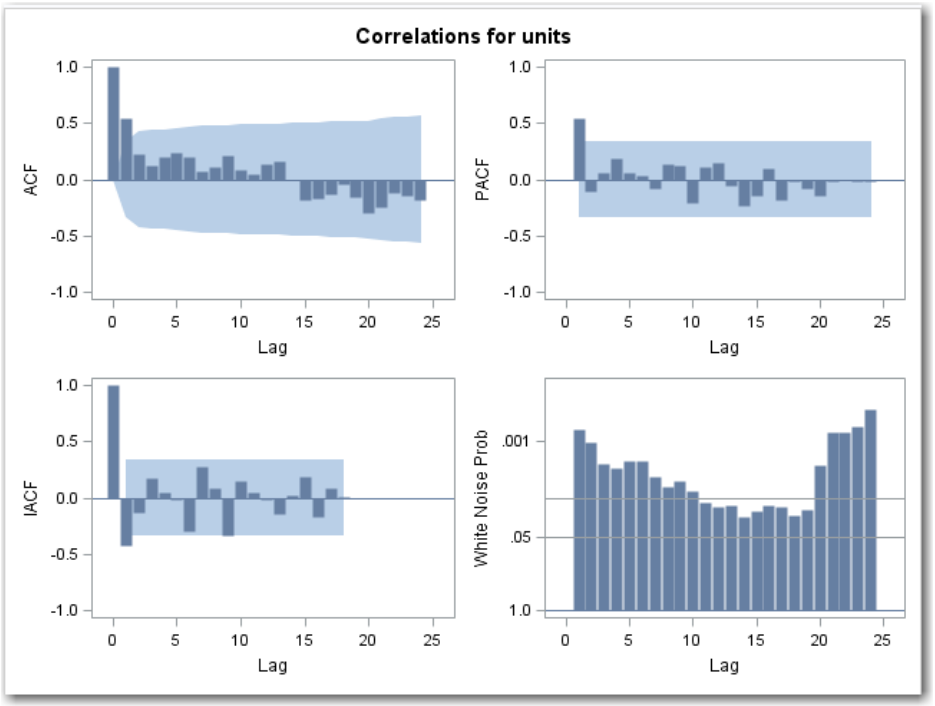


The trend plot is based on a centered, moving-average representation of the series. A strong and linear increase in average sales per month is shown.



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The correlation panel shows that the average **units** series is not white noise. The additional diagnostics provide information that can be used to select autoregressive and moving average orders for ARIMA models. (More information about these is provided in the "ARIMAX Models" chapter.)



A portion of the **decomp_total** data set is shown below. There are 12 unique values of the seasonal components. The peak month, July, is approximately 15% above the annual average. The Seasonally Adjusted series is derived by dividing the **units** (Original) series by the Seasonal Component series.

date	Seasonal Index	Original Series	Trend-Cycle Component	Seasonal Component	Irregular Component	Seasonally Adjusted Series
JAN2004	1	1566.44	.	1.00888	.	1552.65
FEB2004	2	1608.07	.	1.06163	.	1514.72
MAR2004	3	1725.09	.	0.90875	.	1898.30
APR2004	4	1653.51	.	1.00518	.	1644.99
MAY2004	5	1424.55	.	0.95608	.	1489.98
JUN2004	6	1507.31	.	1.02105	.	1476.23
JUL2004	7	1802.05	1635.47	1.14892	0.95903	1568.47
AUG2004	8	1978.14	1642.56	1.12292	1.07247	1761.60
SEP2004	9	1776.46	1633.32	1.06845	1.01796	1662.64
OCT2004	10	1579.73	1625.86	0.99908	0.97252	1581.19
NOV2004	11	1404.00	1644.62	0.82108	1.03972	1709.95

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date	Seasonal Index	Original Series	Trend-Cycle Component	Seasonal Component	Irregular Component	Seasonally Adjusted Series
DEC2004	12	1539.23	1670.01	0.87796	1.04980	1753.18
JAN2005	1	1688.65	1693.65	1.00888	0.98827	1673.79
FEB2005	2	1655.89	1702.78	1.06163	0.91601	1559.76
MAR2005	3	1455.50	1705.19	0.90875	0.93928	1601.64

End of Demonstration

Time Series in SAS Viya Model Studio

Data Dictionary

You will work with a transactional data set named **lookingglass_forecast**. The data consist of information from a telecommunications group whose goal is to predict sales.

The data set is already accumulated into a time series with **Txn_Month** as the time variable. Potential predictor variables of the continuous response **sale** are **price**, **discount**, and **cost**. The data set also contains the attribute variables that will be used in hierarchical forecasting, **productline** and **productname**.

Variables in LOOKINGGLASS_FORECAST

Variable	Type	Length	Format	Informat	Label
Txn_Month	Num	8	MMDDYY10.	DATE15.	Transaction Date (Month)
productline	Char	8			Name of product line
productname	Char	12			Product name
sale	Num	8			Unit Sale
price	Num	8			Unit Price
discount	Num	8			Price Discount
cost	Num	8			Unit Cost

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Setting up a Visual Forecasting Project

In this section, you create a new forecasting project in Model Studio, **baseline sales forecasts**, and load data into the project. The **baseline sales forecasts** project is used throughout the workshop.

1. Navigate to the upper left corner in SAS Viya Landing Page, click **Applications** > **Build Models**. This takes you to Model Studio. Click on the **New Project** button.

   Model Studio is an integrated visual environment that provides a suite of analytic tools to facilitate end-to-end data mining, text, and forecast analysis. The tools in Model Studio are designed to take advantage of SAS Viya programming and cloud processing environments to deliver and distribute the results of the analysis, such as champion models, score code, and results. It does all this fast.

   Note: If this is your first session, there will be no existing projects unless projects were set up for you. (If projects already exist, the **New Project** button is available in the upper right corner.)

2. Name your project **baseline sales forecasts**.

   Note: Naming your project something relevant and adding a reasonably detailed description of the project is considered a forecasting best practice.

3. For **Type**, select **Forecasting**.

   There are three types: Data Mining and Machine Learning, Forecasting, and Text Analytics. This workshop will only deal with Forecasting.

Once you select a type and you create your project, you can't change the Forecasting type. So, you would have to reopen a new project. Make sure you've got the right type before you save the project at the bottom of this window.

4. For **Template**, use the drop-down menu or the Browse button and choose **Auto-forecasting**.
5. For **Data Source**, click **Browse** to select the modeling data source.
6. Make sure the Show area says **Available** to view in-memory tables that are available for model building. Scroll and find the **LOOKINGGLASS_FORECAST** in-memory table. Alternatively, you can click on the search bar and type lookingglass and click the magnifying glass icon or hit Enter.
7. Select **LOOKINGGLASS_FORECAST**, click **OK**, and then **Save** to create the new project. The project now appears.
8. Ensure that the project's **Data** tab is selected to assign variable roles.

   A Note on Variable Assignment:

   - Individual variables can be selected for role assignment by either clicking the variable name or by selecting the corresponding check box.

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- Individual variables are **deselected** after their role is assigned by either **clearing their check box** or **selecting another variable's name**.
- More than one variable can be selected at the same time using the check boxes.
- Because selecting a variable using the check box does not deselect others, it is easy for new users to **inadvertently re-assign** variable roles. Taking a few minutes to get comfortable with the variable selection functionality is considered a best practice for using the software.

9. The **Txn_Month** variable is assigned to the role of Time for the project.

   Note: Other time intervals are available by selecting the down arrow next to **Month**. The time interval combined with the Multiplier and Shift options indicates that the desired interval of the time series data is one month and that the 12-month annual cycle starts in January. These options can be changed to modify the time index if it is appropriate for your data.

10. **Sale** is the target for the analysis. Click **sale** in the middle variables list panel. In the right property panel, select **Dependent**.

    Note: Missing interpretation options enable the user to interpret, or impute, values for embedded missing values in the series. By default, embedded missing values have no value assigned to them.

11. Deselect **sale**. Assign **productline** and **productname,** in that order, to the **BY Variable** role.
12. Set Reconciliation level to **productline**.
13. Additional variables will be assigned to roles. **Price**, **discount**, and **cost** can be useful as explanatory variables in subsequent analyses. Select these three variables, where the order does not matter, and change their roles to **Independent**.

    Note: For each of these variables, accumulation is accomplished by averaging observed values in each month.

14. Change **Usage in system-generated models** to **Try to Use**.

If I select Force to use, then each of these three variables will be used in every one of the models for every one of the series. I don't want to do that. Let's see some other options-- Try to use and Use if significant.

Try to Use will test each of the variables in each of the series. So, for each model, for each of the 918 series, in the dataset each of the variables will be tested to see if they're statistically significant in the model, and whether they benefit the model with respect to a fit statistic, such as Akaike's information criterion. So, there are two criteria. If a variable passes both of those tests, they'll be used in the model. If the variable doesn't, it won't be used in the model.

Try to Use is slightly different from Use if significant. Use if significant only tests to see if the variable is statistically significant and doesn't check to see if it improves the model's fit statistic.

So I'm going to use Try to use. Now my data are ready to start my pipelines.

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Creating Pipelines - Autoforecasting

In this section, you perform basic forecasting with a pipeline.

1. Navigate to the **Pipelines** tab.

   The Auto-Forecasting template is the default pipeline template for Visual Forecasting. It consists of the essential steps in a forecasting analysis:

   - accumulates the data into time series
   - automatically identifies, estimates, and selects forecast models for the time series
   - assesses forecasting results
   - publishes results for use outside the pipeline

   Note: If the modeling data are hierarchically arranged, the identification, estimation, and selection steps in the default forecasting pipeline are done on series in the base level of the hierarchy.

2. Select **Run Pipeline** in the upper right corner of the workspace.

   Then we see the circles, the empty circles start up. And by the time we see all circles filled with green ink and a white checkmark, then the pipeline will be completed. Now that it's done, let's look at the Auto Forecasting node.

Auto-forecasting Node Results

1. Right-click on the **Auto-forecasting** node and select **Results**.

   Because the Auto-forecasting node is designed to be run with minimal input from the analyst, relatively few options are surfaced for this node. The Auto-forecasting node automatically identifies, estimates, and generates forecasts for the 918 series in the base or product name level of the modeling hierarchy. Most of the forecast models selected for these series are in the ARIMAX family.

   For each series, three families of time series models are considered by default: ARIMAX (ARIMA with exogenous variables), ESM (exponential smoothing models), and IDM(intermittent demand models). The champion model for each series is chosen based on root mean square error. Other selection statistics are available in the Model selection criterion option.

   The MAPE Distribution histogram is in the upper left-hand corner. The distribution of Mean Absolute Percent Error (MAPE) for forecasts in the product name level of the hierarchy can be used to compare the accuracy of different forecast models. Each of the bars represents the proportion of the series that have a specific range of MAPE values. In general, smaller values of MAPE imply greater accuracy.

   The Model Family histogram is in the upper right. Among each of these 918 series, we can see what percentage of each model family was selected as the best across all the time series.

   The Model Type chart, located in the lower left, summarizes systematic variation found in the identification process. This chart shows information about the selected models across all the time series.

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What percentage of the models has a trend component, a seasonal component, or an predictor variable used? They can be overlapping series; therefore, we can see that the percentage is summed to over 100%.

The Execution Summary, located in the lower right, provides information about results that are potentially problematic, anomalous, or both. We can see there were 918 series. There weren't any series that failed for forecasting. There were only six series with forecasts equal to zero, meaning in the forecast range, the forecasts were all zero. Then there is a lot of summary information about the number of series that had flat forecasts. Flat forecasts means that in the forecast range, the forecast values were a constant.

2. Click the **Output Data** tab above the MAPE Distribution plot.

   Several output tables are created. You can view them by clicking on them.

   You'll notice that there is a unique product line, product name combination for each line. So that's a unique series identified by its product line and product name. And notice that we have multiple lines of data in this data set, each for a different month. So, if I scroll down, I can see that the Time ID. And once we get to 2017, we see the actual values is missing. What that means is that this is the forecast horizon. The forecast horizon, of course, doesn't have any actual values, but it can have predicted values and so on. So that's information you might want to obtain from that forecast table.

3. Click the **OUTMODELINFO** data source to open it.

   For each series, the selected model is named, and attributes of the model are displayed.

   Once again, we see information for every product line, product name combination. In other words, every series. And in this data set, we see the name of the model, or the label for the model, is the type of model that was chosen as the champion model for that series. So, for Line01, Product01, that series, it was an ARIMA model with regression parameters. It is under the ARIMA family. There were no dependent variables here. And we can get information about whether there are seasonal components, whether there are trend components, whether there are inputs presence, and so on.

   So that's information for the Champion model. If I want to see what the competitors were, we can click on the OUTSELECT Data Source. And now you'll see there are three lines for each one of the series. So Line01, Product01 is three different lines. And you can see which of the models were under consideration by looking at the Model column. And then the next column for Selected Status, you can see that the selected row, the selected model for this series, as we'd seen before, was the ARIMA model with regression parameters. And you can see why that happened by scrolling farther to the right, each one of these has fit statistics and accuracy statistics calculated. So, if we looked at the Mean Absolute Percent Error column, for those first three rows, you could see why that model won the competition for that series.

4. Close the **Results** window.

5. Right-click and open the results of the **Model Comparison** node.

   The Champion Model is the Auto-forecasting model, which is the only one included in the pipeline. WMAE and WMAPE are weighted sums of the MAE and MAPE values across all series. WMAPE and WMAE represent average performance of all the models in a modeling node.

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Note: For the WMAPE and WMAE, the final computation is based on weighted measurements from each time series, where more weight is given to time series with a higher average of the dependent variable.

6. Close the **Results** window.

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